

TM Forum Specification

Service Qualification Management User Guide

TMF645

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Table of Contents

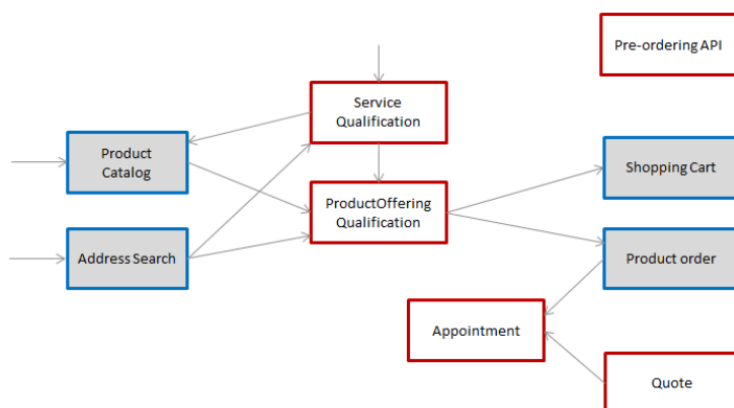
NOTICE	i
Introduction	1
Lifecycle	3
Sample Use Cases	4
Support of polymorphism and extension patterns	5
RESOURCE MODEL	6
Managed Entity and Task Resource Models	6
QueryServiceQualification resource	6
CheckServiceQualification resource	60
Notification Resource Models	74
QueryServiceQualification	75
CheckServiceQualification	79
API OPERATIONS	85
Operations on QueryServiceQualification	85
List or find QueryServiceQualification objects	85
Retrieves a QueryServiceQualification by ID	88
Initiate a task for QueryServiceQualification	90
Updates partially a QueryServiceQualification	93
Deletes a QueryServiceQualification	99
Operations on CheckServiceQualification	100
Retrieves a CheckServiceQualification by ID	100
List or find CheckServiceQualification objects	102
Initiate a task for CheckServiceQualification	105
Updates partially a CheckServiceQualification	109
Deletes a CheckServiceQualification	116
API NOTIFICATIONS	118
Register listener	118
Unregister listener	119
Publish Event to listener	119
Acknowledgements	121
Release History	121
Contributors to Document	121

Introduction

The following document is the specification of the REST API for ServiceQualification management. It includes the model definition as well as all available operations.

In the Open Digital Economy where multiple actors (SDPs, CSPs, ...) may be involved with the delivery of an end-to end service, those actors need to collaborate and interact with the customer as needed.

Following diagram shows ServiceQualification API within pre-ordering API package:



Service Qualification API is one of Pre-Ordering Management API Family. Service Qualification API goal is to provide service technical eligibility in the context of the interaction. This API allows to:

- Request a specific service (of an array of service) eligibility providing as an input configured service. The resource `checkServiceQualification` is used for this purpose. The requester will provide an array of service qualification item and expect to have a qualification result for each one of them. In the response, the API will provide qualification result but also:
 - Eligibility unavailability reason (if any)
 - Alternate service proposal (if any).
- Retrieve available services based on the `searchCriteria` provided in the request - The primary use case involves supplying a service category and a location as `searchCriteria`. In response, the requester receives:
 - The original `searchCriteria` provided.
 - An array of `ServiceQualificationItems`, each representing an available service that matches the criteria.

Note: If no services match the criteria, the response will still include the original

searchCriteria, but no error will be triggered. The serviceQualificationItem array will simply be empty.

Because the API allows describing service from the inventory, it is also possible to use it to retrieve service available to complete existing service for both resources.

This API call does not modify the service in the inventory. Specifically, it does not change the service state (there is no transition to the eligibilityChecked state)

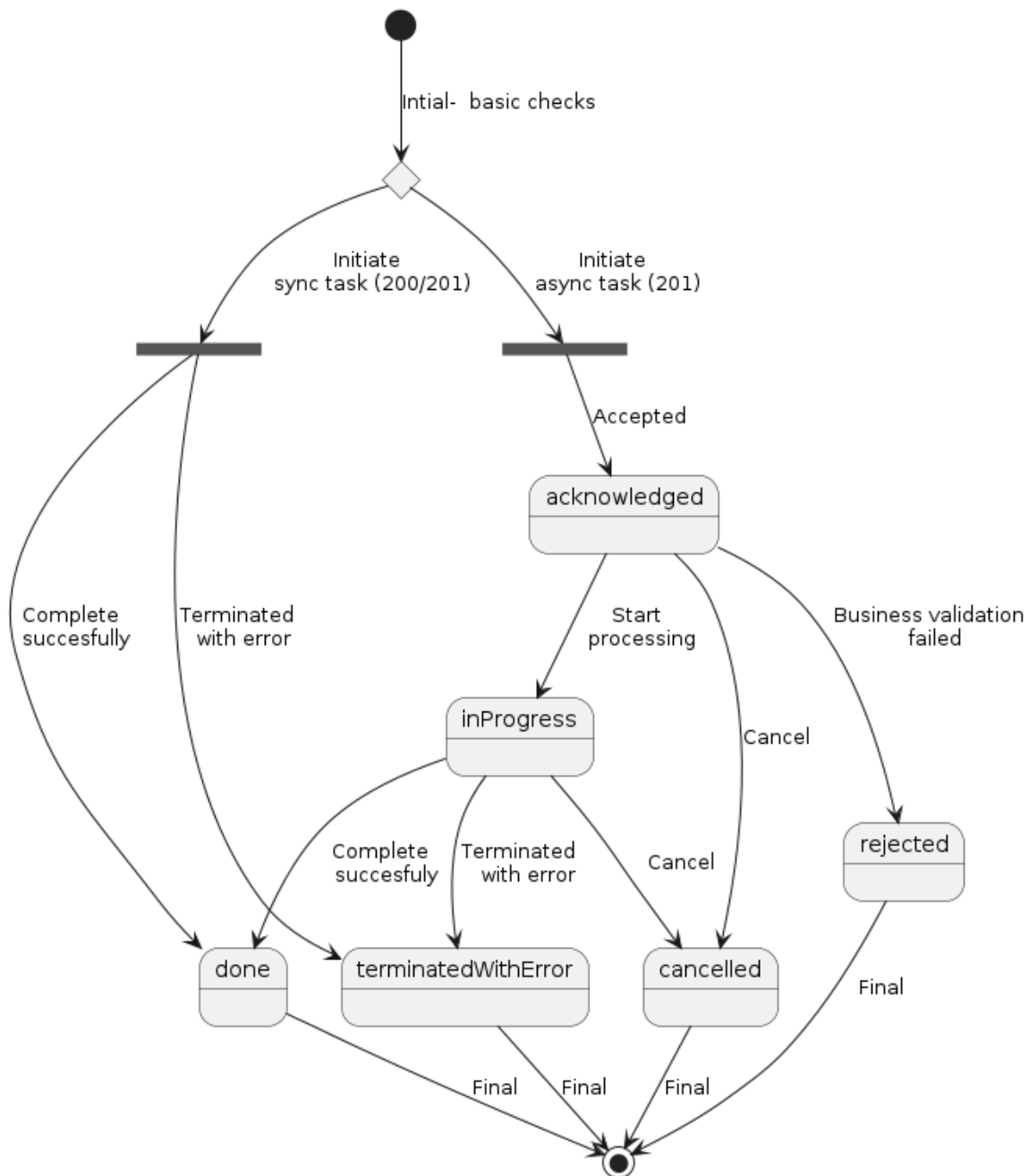
In the request, for both resources, the following attributes could be filled:

- service, serviceSpecification, category or service characteristics will be indicated (at least one is mandatory),
- relatedParty, for which the qualification must be done,
- place, where the customer wants his service,
- expected qualification date.

Lifecycle

The state machine specifying the typical state change transitions is provided below. This is standard task based resource lifecycle:

TMF Task Lifecycle



Sample Use Cases

To provide some use-cases:

For checkServiceQualification:

- Check if we can deliver a 4k TV service at a specified address
- Check if we can deliver new IP TV service at the same location of an existing and active Access service (i.e. without explicitly specifying address but using service relationship)
- Check if we can upgrade the download speed of an existing and active service from 100 Mb/s to 600Mb/s

For queryServiceQualification:

- Retrieve a list of service specification – from a service category - available at a given place
- Retrieve a list of service specification connectable to an existing service

Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a `CheckServiceQualificationTypeA` and `CheckServiceQualificationTypeB` inheriting properties from the abstract `CheckServiceQualification` entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines, Part 2 (TMF630).

The `@type` attribute provides a way to represent the actual class type of an entity. For example, within a list of `CheckServiceQualification` instances some may be instances of `CheckServiceQualificationTypeA` where other could be instances of `CheckServiceQualificationTypeB`. The `@type` gives this information. All resources and sub-resources of this API have a `@type` attributes that can be provided when this is useful.

The `@referredType` can be used within reference entities (like for instance an `ServiceSpecificationRef` object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the `@type`, when used, denotes the class type of the reference itself, such as `ServiceSpecificationRef`, and not the class type of the referred object. However since reference classes are rarely sub-classed, `@type` is generally not useful in reference objects.

The `@schemaLocation` property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected characteristics of an entity.

The `@baseType` attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

RESOURCE MODEL

Managed Entity and Task Resource Models

QueryServiceQualification resource

QueryServiceQualification is used to retrieve a list of services that are technically available in the context of the interaction (place, party, service characteristics, ...).

Resource model

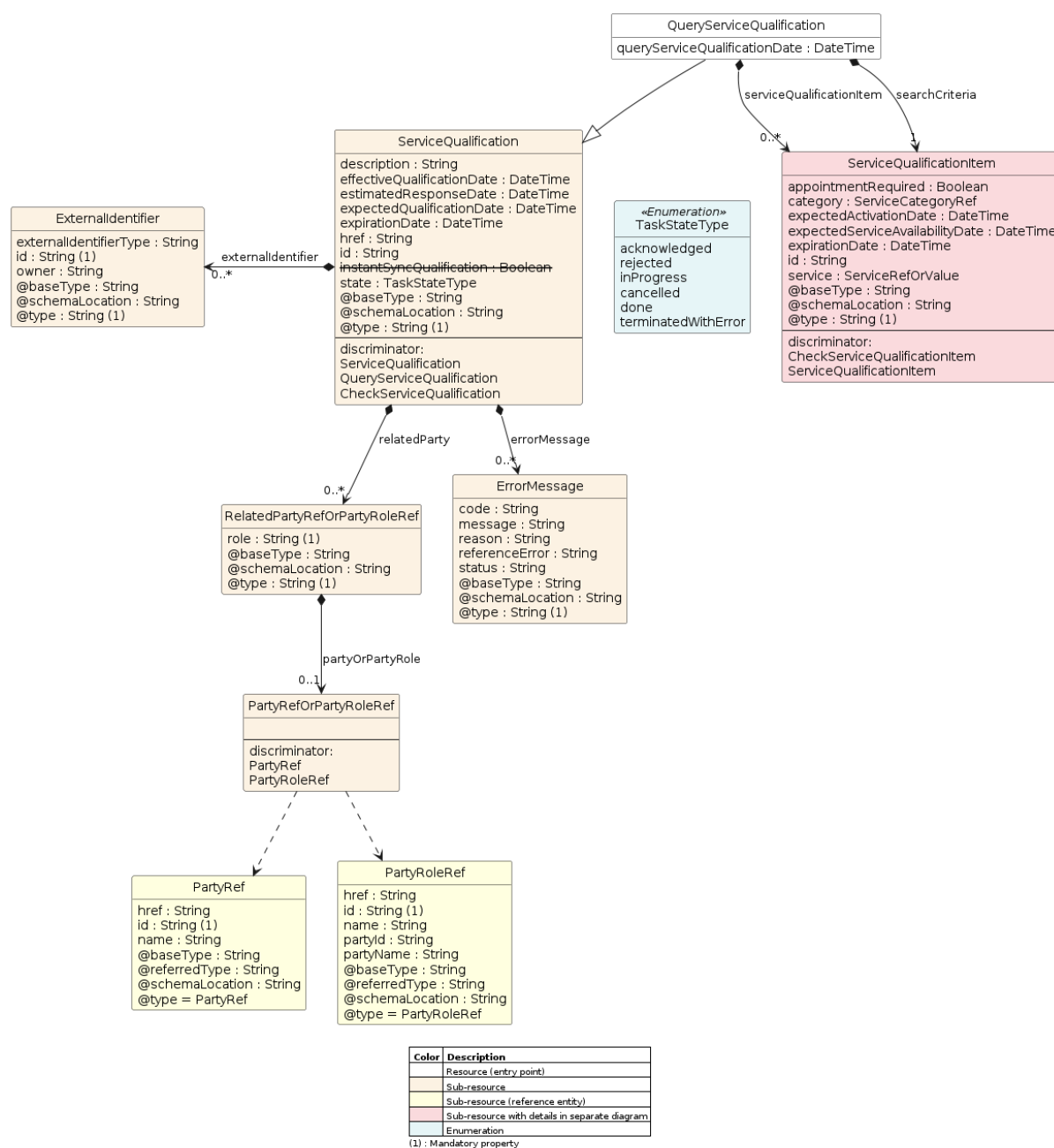


Figure 1 - QueryServiceQualification

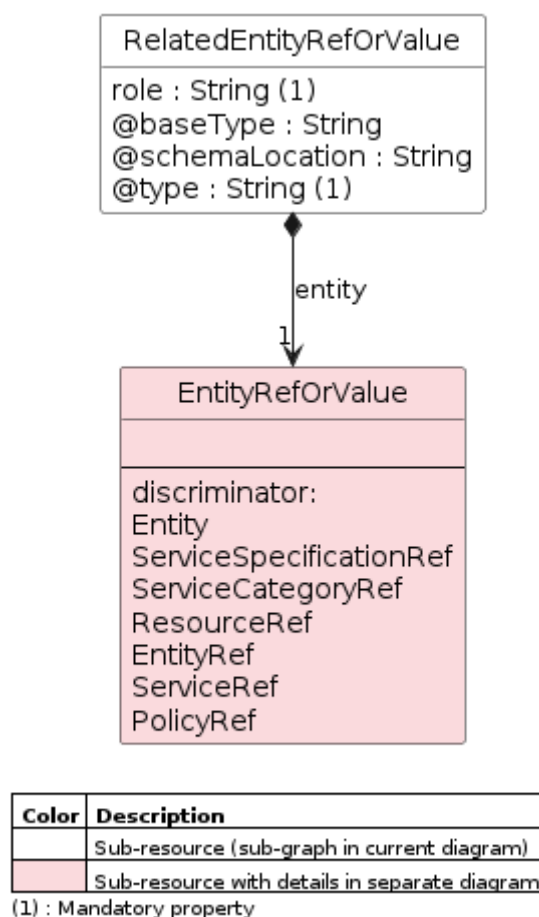


Figure 2 - RelatedEntityRefOrValue

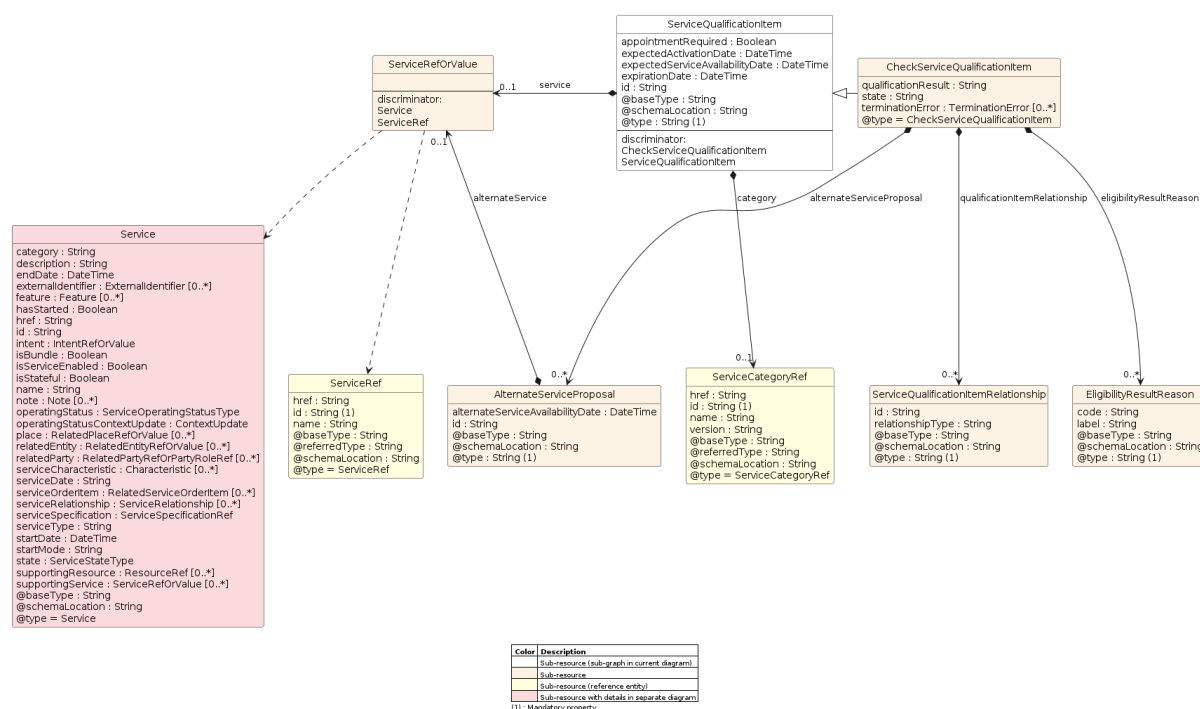


Figure 3 - ServiceQualificationItem

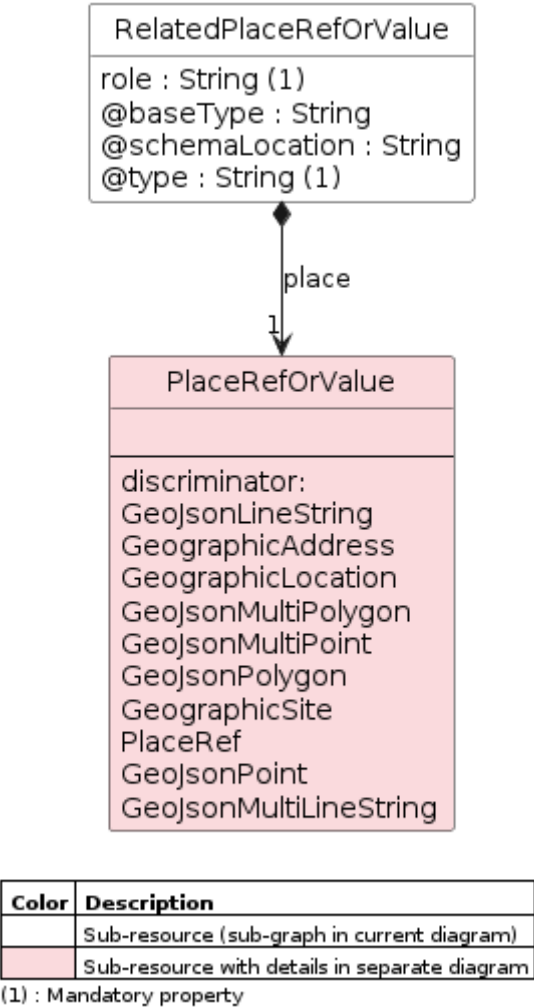


Figure 4 - RelatedPlaceRefOrValue

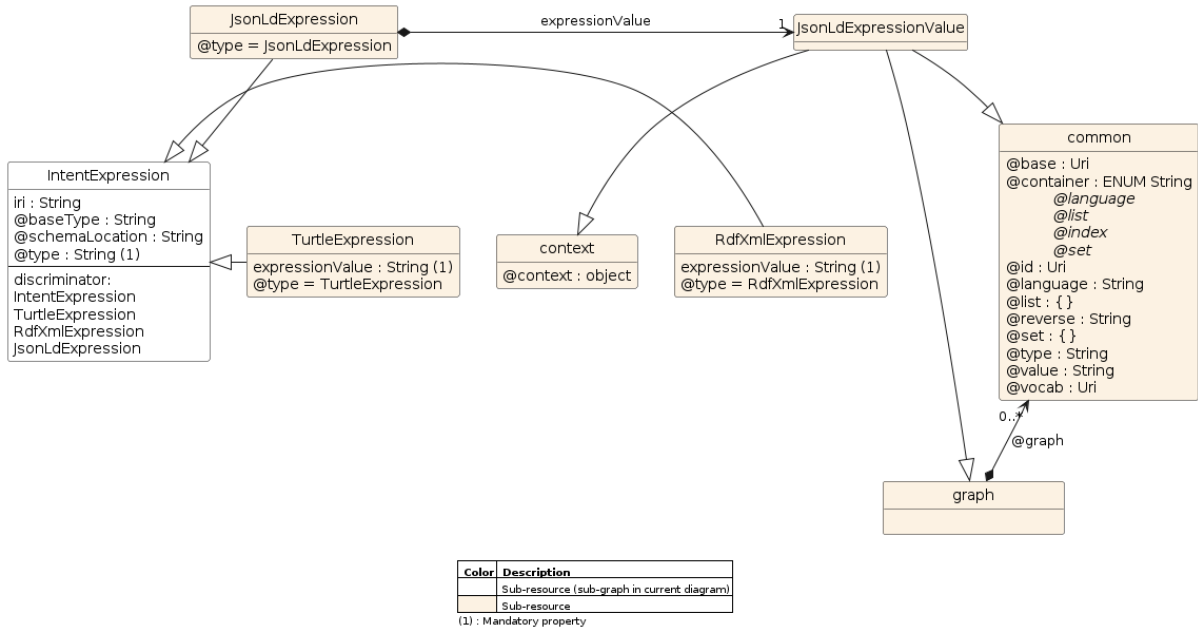


Figure 5 - IntentExpression

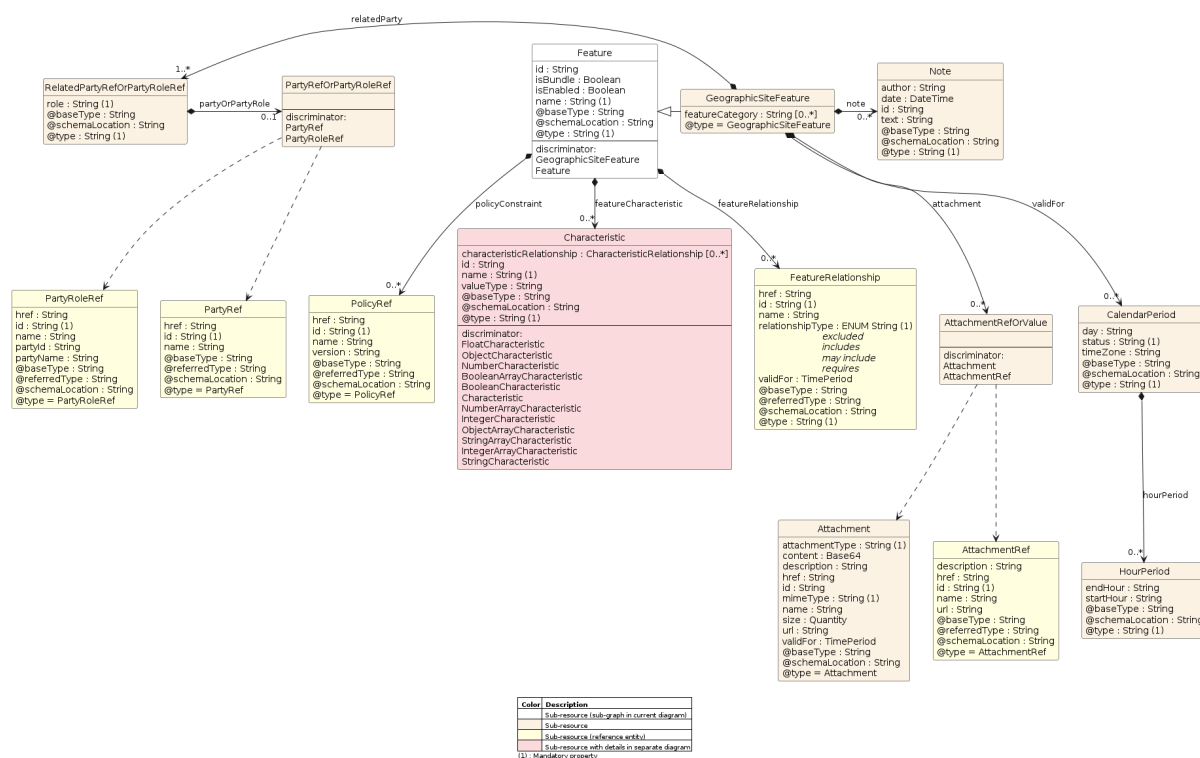


Figure 6 - Feature

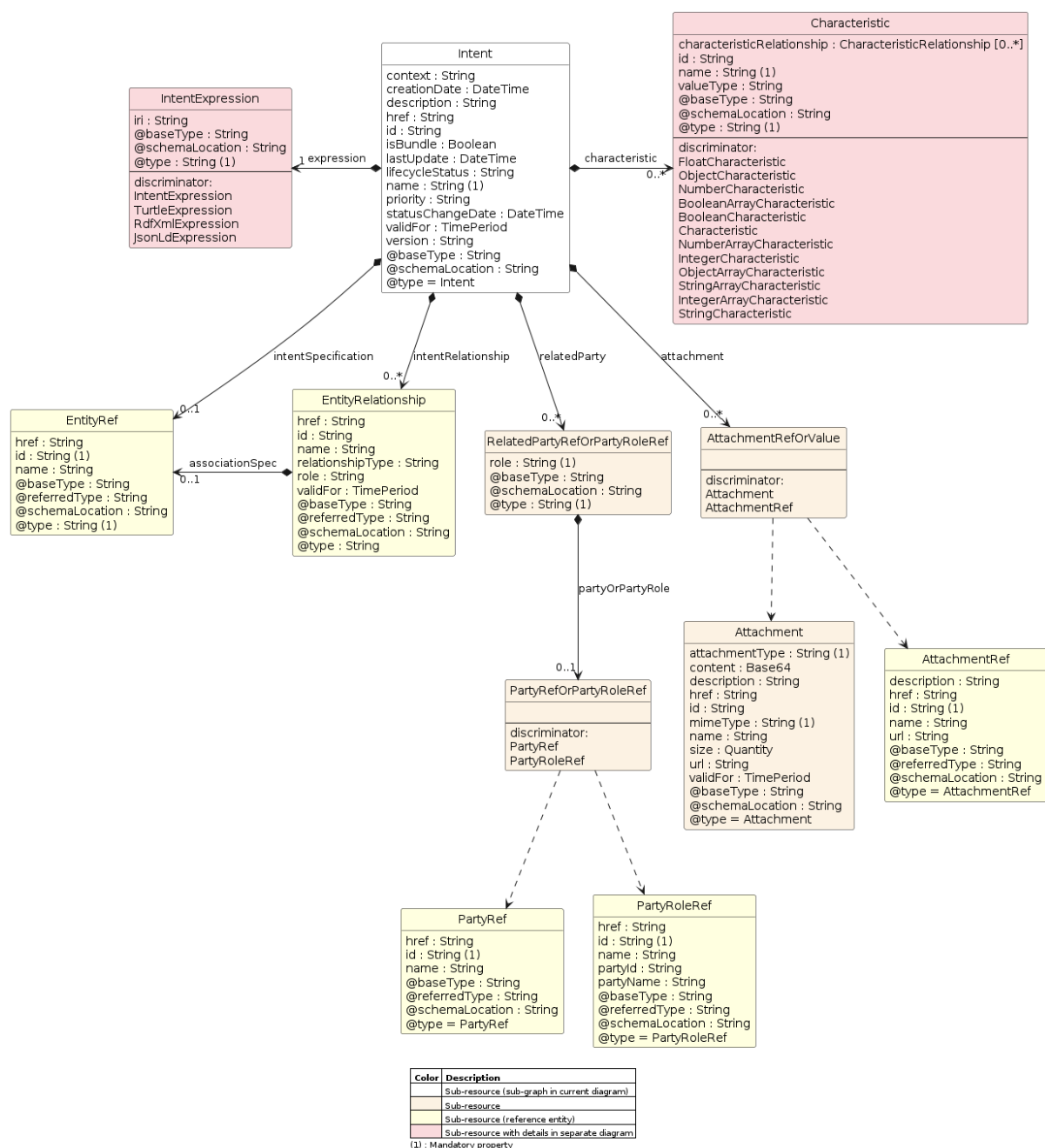


Figure 7 - Intent

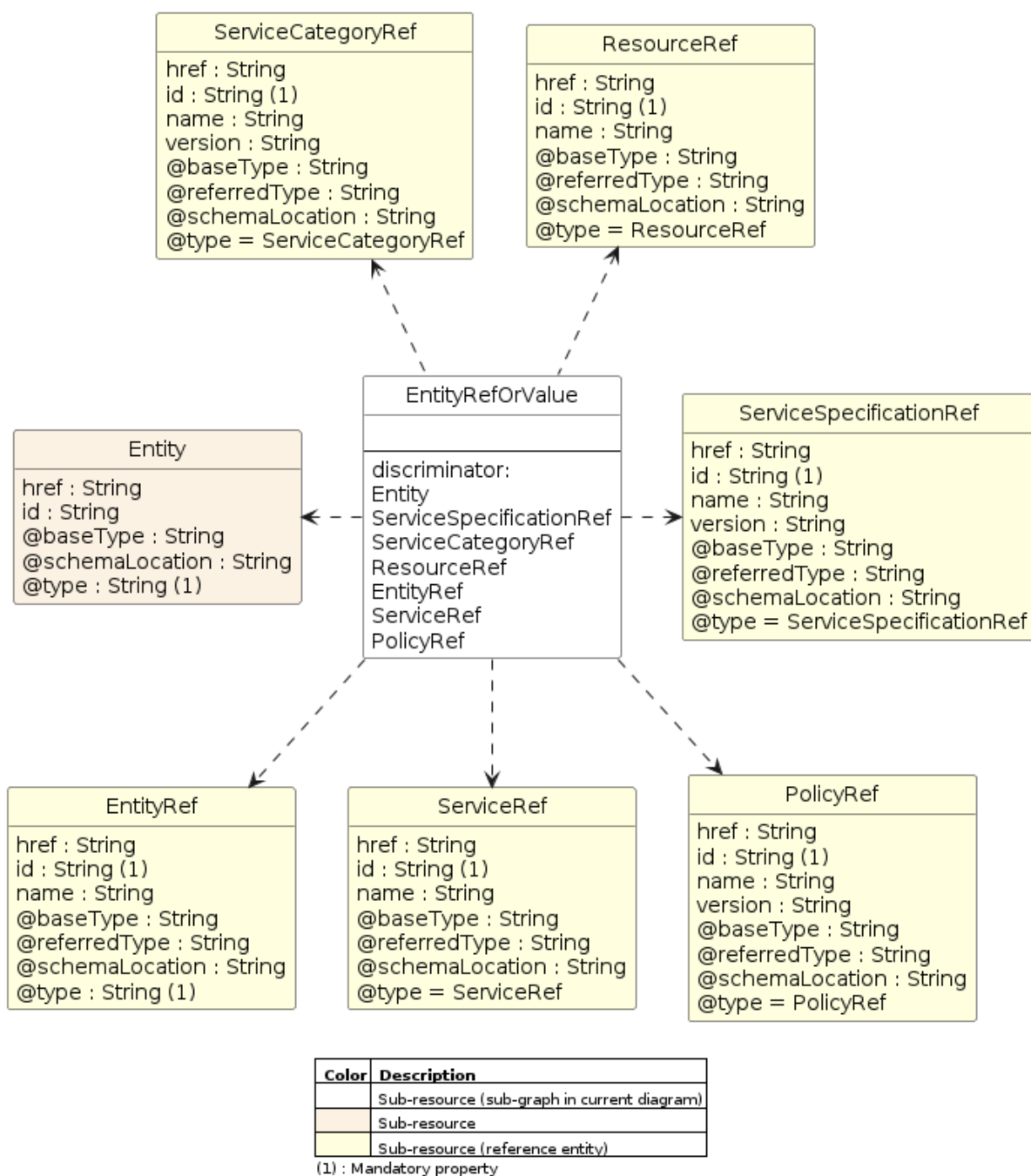


Figure 8 - EntityRefOrValue

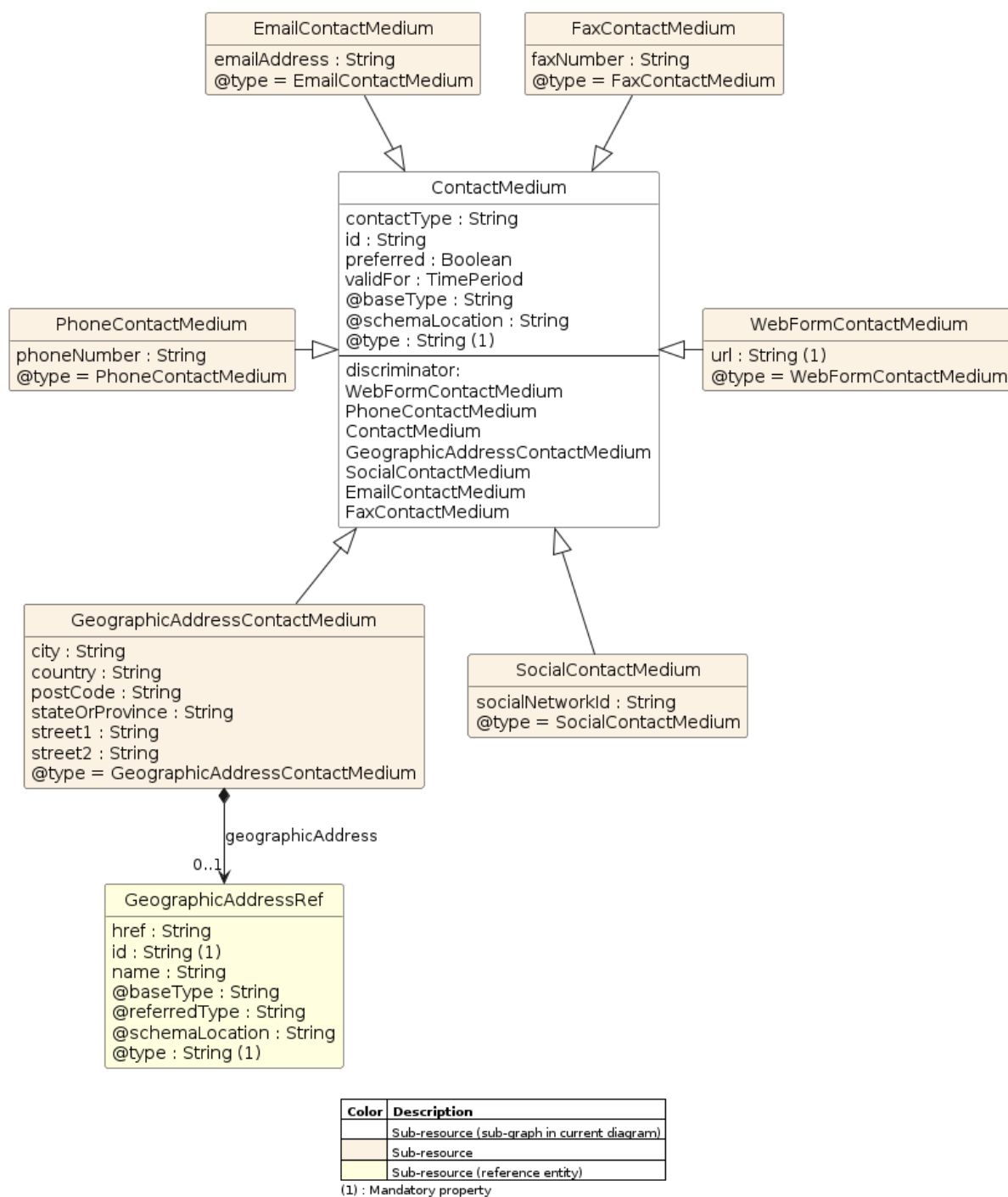


Figure 9 - ContactMedium

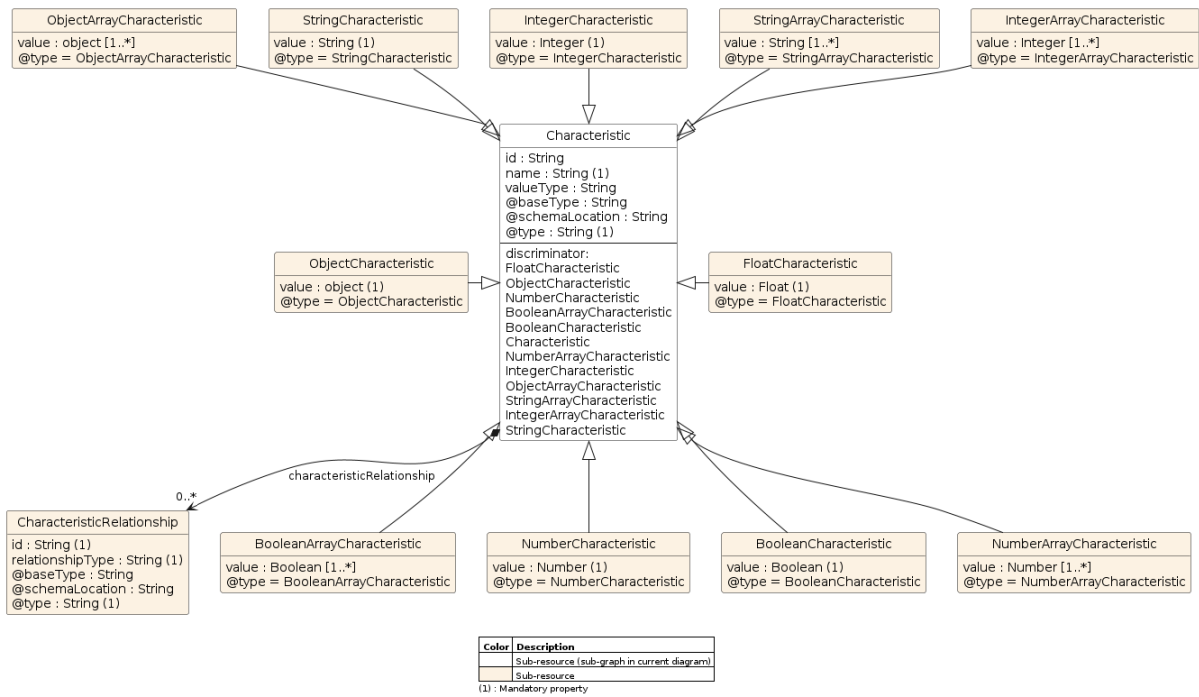


Figure 10 - Characteristic

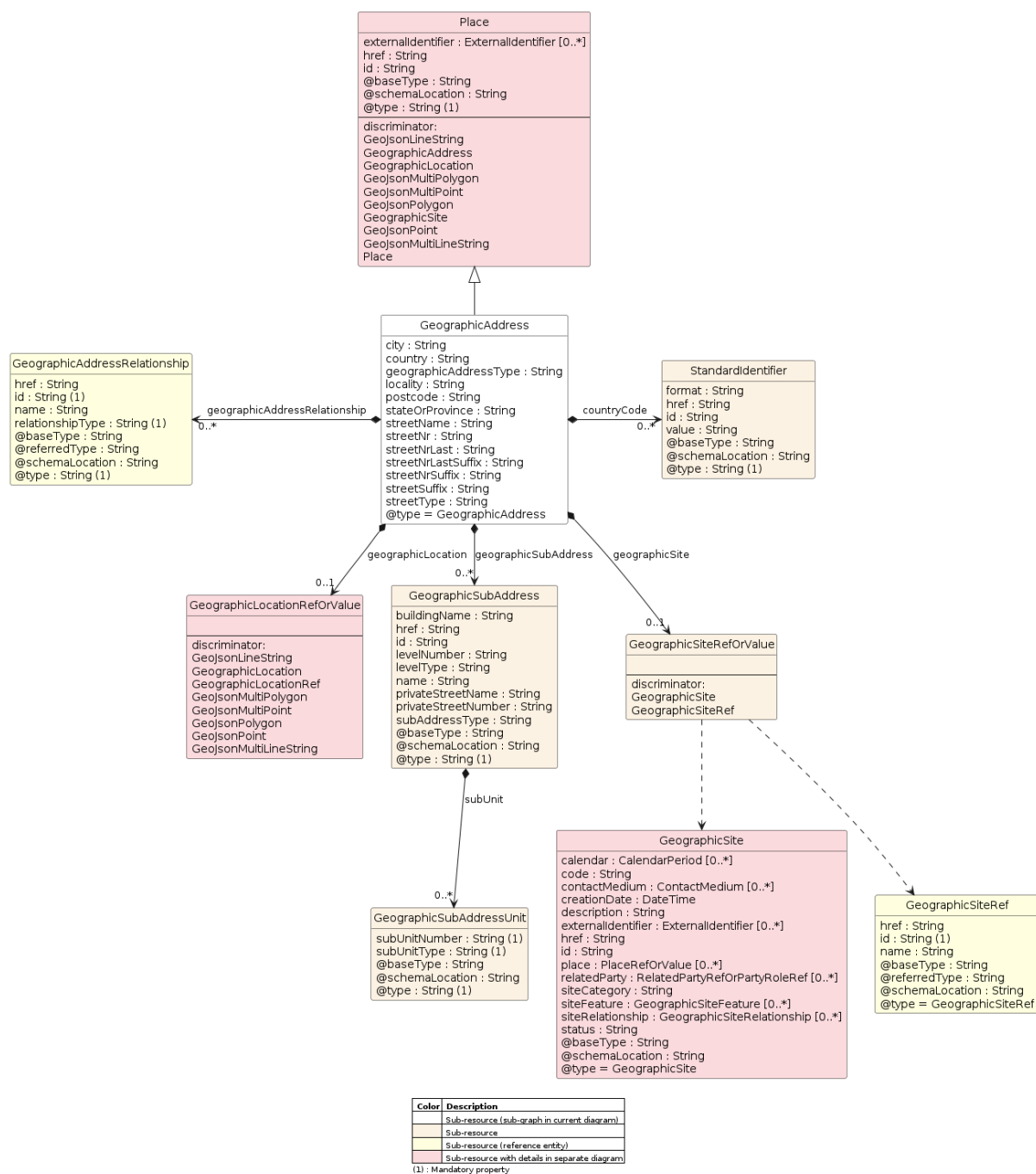


Figure 11 - GeographicAddress

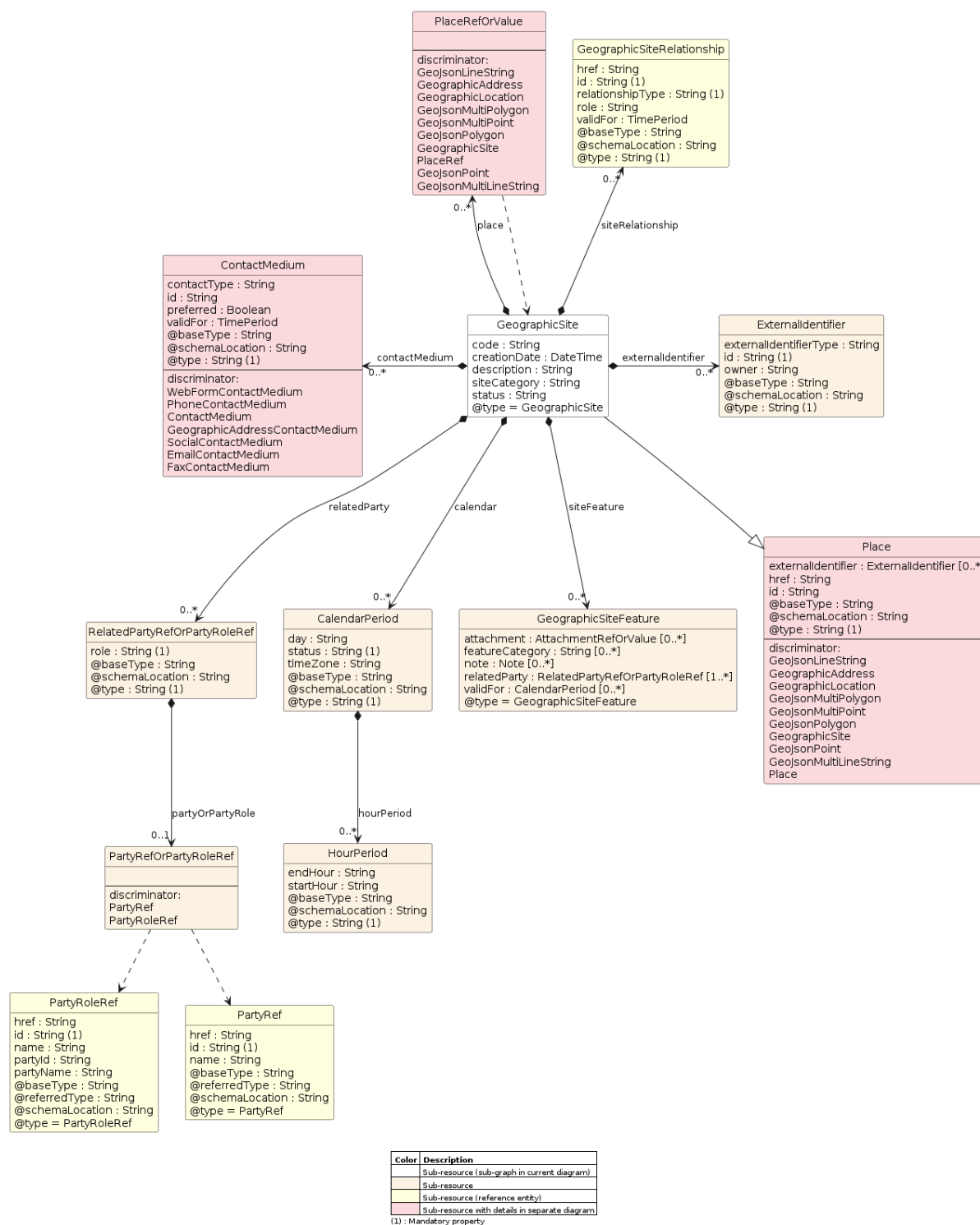
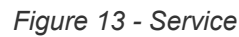


Figure 12 - GeographicSite



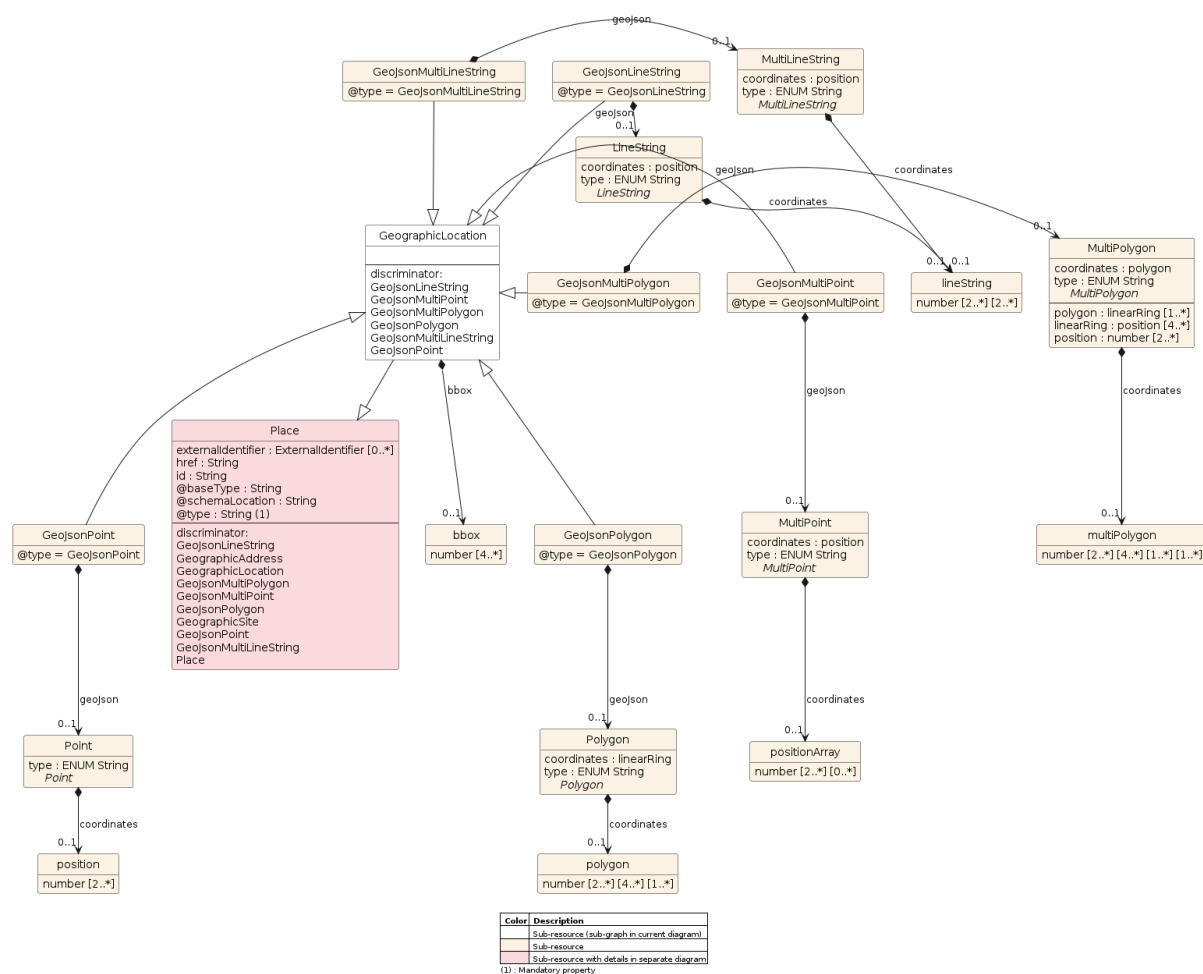


Figure 14 - GeographicLocation

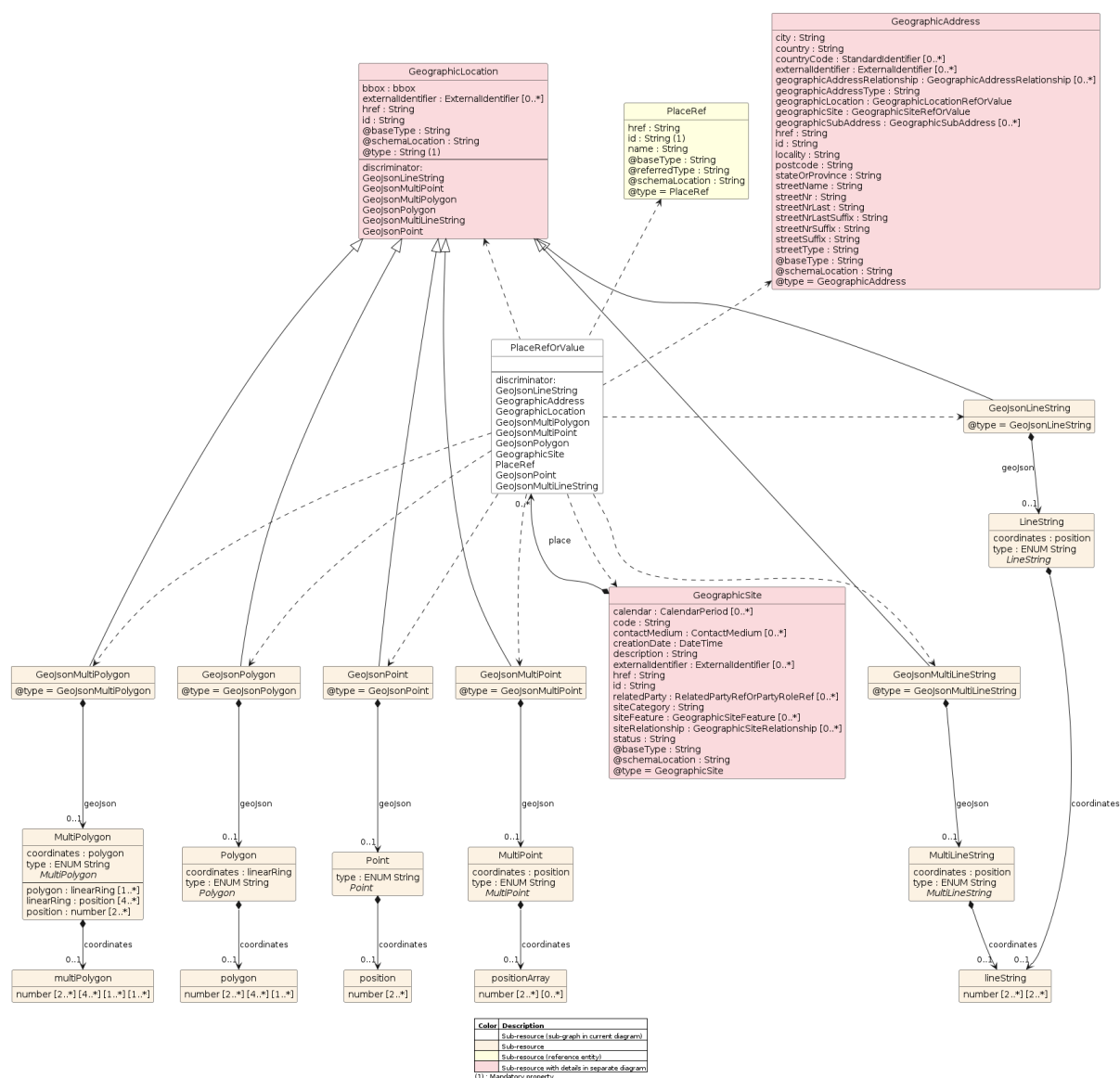


Figure 15 - PlaceRefOrValue

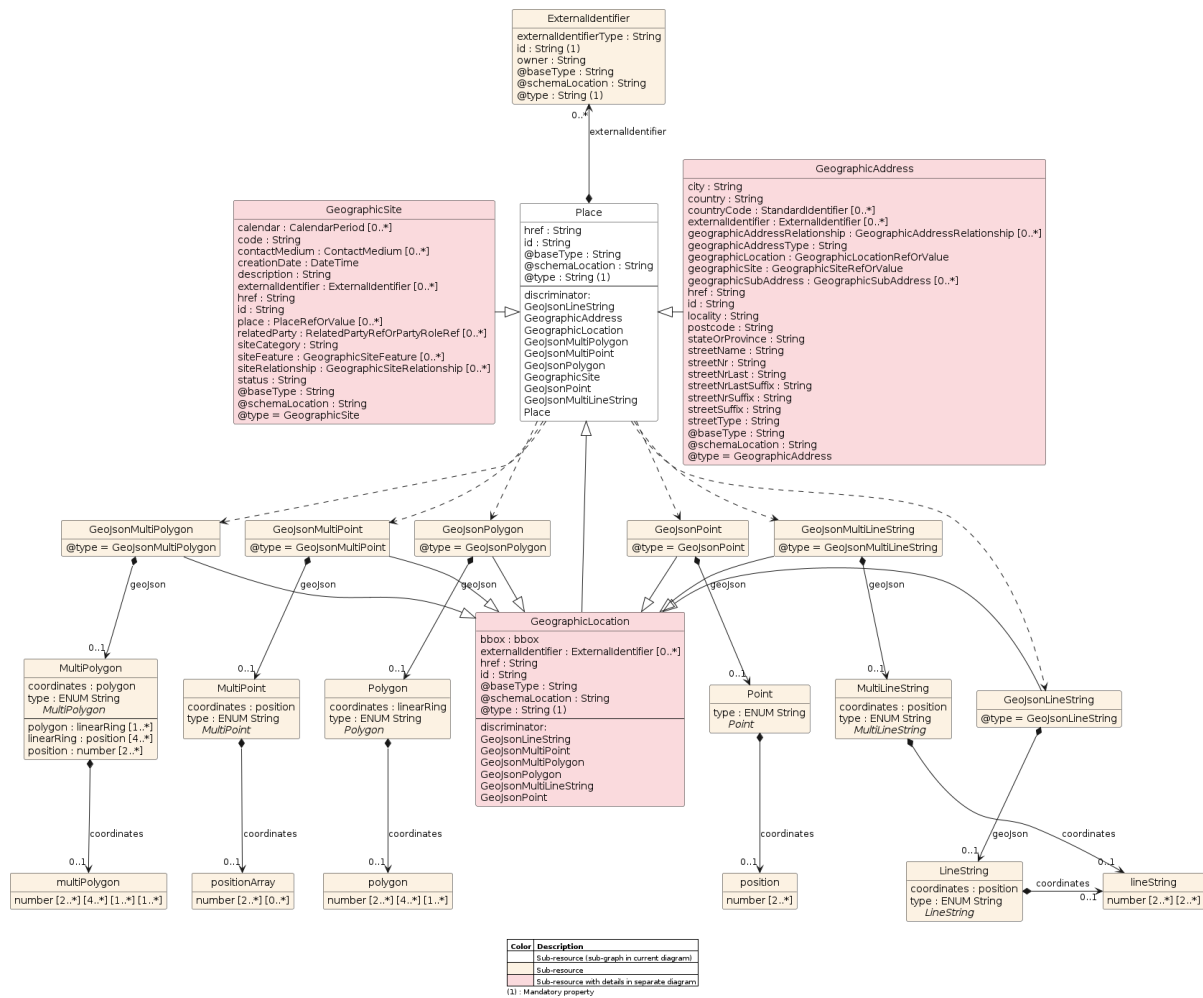


Figure 16 - Place

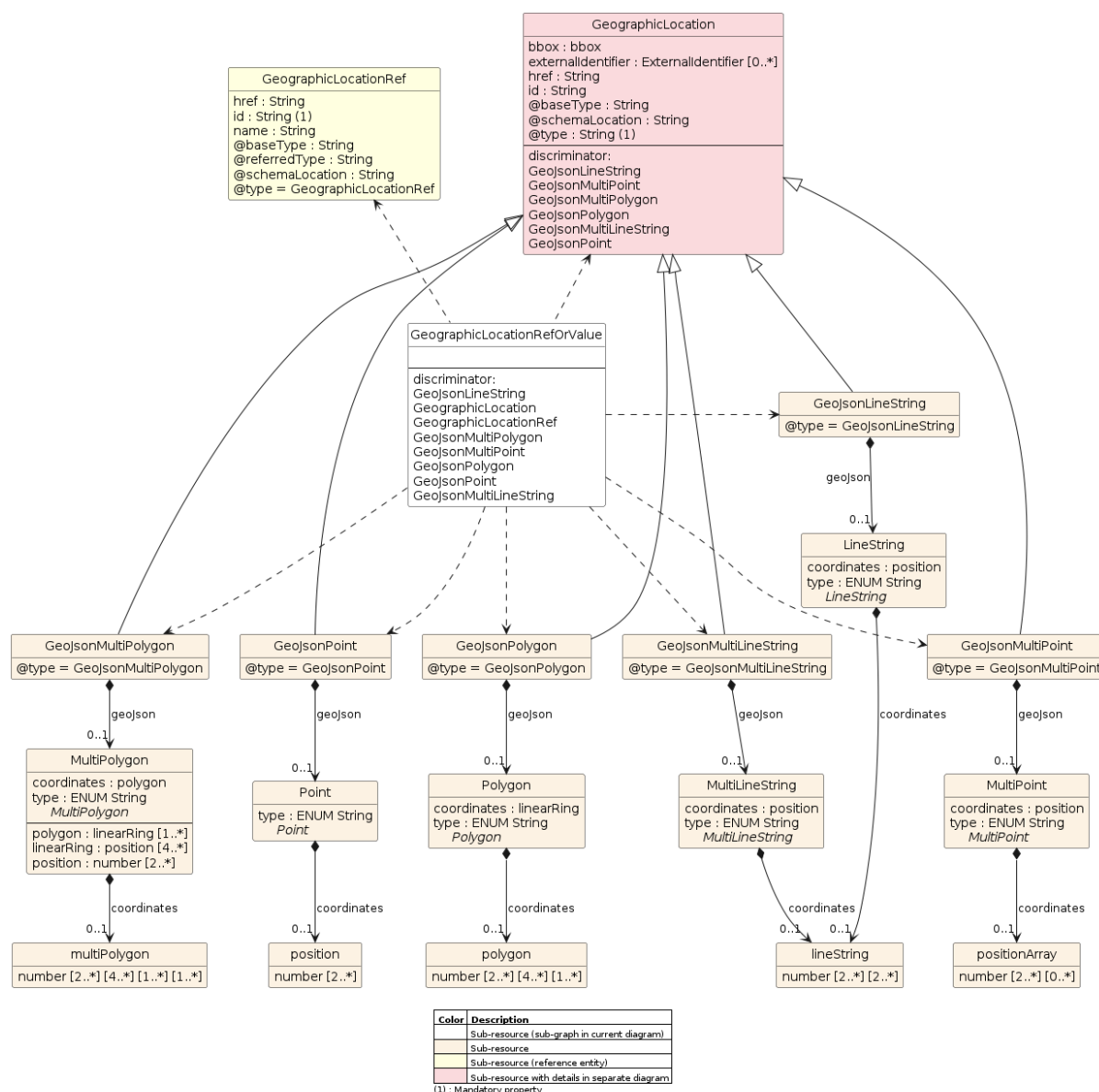


Figure 17 - GeographicLocationRefOrValue

Field descriptions

QueryServiceQualification fields

description	A String. Description of the serviceQualification.
effectiveQualificationDate	A DateTime. Effective date to serviceQualification completion.
errorMessage	An ErrorMessage. A list of error message contains the error that caused the termination of ServiceQualification task.
estimatedResponseDate	A DateTime. A Date by when the provider estimates supplying a response to the qualification request.
expectedQualificationDate	A DateTime. A date (DateTime). Deadline date when the requester expected a qualification answer.
expirationDate	A DateTime. Date when the qualification response expires.

externalIdentifier	An ExternalIdentifier. A list of external identifiers associated with this resource.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
instantSyncQualification	DEPRECATED A Boolean. A previous indicator which when the value is "true" means that requester expects to get qualification result immediately in the response. If the indicator is true (or by default now without it) then the response code of 200 indicates the operation is successful otherwise a task is created with a response 201. For asynchronous, this attribute is now replaced by the header request "Expect: 202-Accepted".
queryServiceQualificationDate	A DateTime. Date when the serviceQualification was submitted.
relatedParty	A RelatedPartyRefOrPartyRoleRef. A list of related party references, defines party or party role linked to this request.
searchCriteria	A ServiceQualificationItem. A ServiceQualificationItem relates to a specific service being checked in a qualification operation. ServiceQualificationItem can be instantiated as * CheckServiceQualificationItem
serviceQualificationItem	A ServiceQualificationItem. A list of service qualification items. ServiceQualificationItem can be instantiated as * CheckServiceQualificationItem
state	A TaskStateType. Possible values for the state of a task. ENUMERATED with values: * acknowledged * rejected * inProgress * cancelled * done * terminatedWithError
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ErrorMessage sub-resource fields

code	A String. Error code.
message	A String. More details and corrective actions related to the error.
reason	A String. Explanation of the reason for the error.
referenceError	A String. URI of documentation describing the error.
status	A String. Error code extension like sys-ABC-2001.

@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ExternalIdentifier sub-resource fields

externalIdentifierType	A String. Type of the identification, typically would be the type of the entity within the external system.
id	A String. Identification of the entity within the external system.
owner	A String. Name of the external system that owns the entity.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyRefOrPartyRoleRef sub-resource fields

partyOrPartyRole	A PartyRefOrPartyRoleRef.
role	A String. Role played by the related party or party role in the context of the specific entity it is linked to. Such as 'initiator', 'customer', 'salesAgent', 'user'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
partyId	A String. The identifier of the engaged party that is linked to the PartyRole object.
partyName	A String. The name of the engaged party that is linked to the PartyRole object.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ServiceQualificationItem sub-resource fields

appointmentRequired	A Boolean. Indicates whether an appointment is required or not.
category	A ServiceCategoryRef. The (service) category resource is used to group service candidates in logical containers. Categories can contain other categories.
expectedActivationDate	A DateTime. The date when the service is expected to be activated.
expectedServiceAvailabilityDate	A DateTime. Date when the requester looks for service availability.
expirationDate	A DateTime. Date when the qualification item response expires.
id	A String. Id of the Service Qualification Item.
service	A ServiceRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Service entity and not the ServiceRefOrValue class itself.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	ServiceQualificationItem can be instantiated as * CheckServiceQualificationItem
alternateServiceProposal	This property is present in subclasses
eligibilityResultReason	This property is present in subclasses
qualificationItemRelationship	This property is present in subclasses

qualificationResult	This property is present in subclasses
state	This property is present in subclasses
terminationError	This property is present in subclasses

ServiceCategoryRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Category version.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Service sub-resource fields

category	A String. Is it a customer facing or resource facing service.
description	A String. Free-text description of the service.
endDate	A DateTime. Date when the service ends.
externalIdentifier	An ExternalIdentifier. A list of external identifiers associated with this service.
feature	A Feature. A list of feature associated with this service. Feature can be instantiated as * GeographicSiteFeature
hasStarted	A Boolean. If TRUE, this Service has already been started.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intent	An IntentRefOrValue. Intent Ref (if Intent already exists) or Value (if Intent be created or its details be presented).
isBundle	A Boolean. If true, the service is a ServiceBundle which regroup a service hierachy. If false, the service is a 'atomic' service (hierachy leaf).
isServiceEnabled	A Boolean. If FALSE and hasStarted is FALSE, this particular Service has NOT been enabled for use - if FALSE and hasStarted is TRUE then the service has failed.
isStateful	A Boolean. If TRUE, this Service can be changed without affecting any other services.

name	A String. Name of the service.
note	A Note. A list of notes made on this service.
operatingStatus	A ServiceOperatingStatusType. Valid values for the Operating status of the service. ENUMERATED with values: * pending * configured * starting * running * degraded * failed * limited * stopping * stopped * unknown
operatingStatusContextUpdate	A ContextUpdate. Holds the date when the status/state last changed and provides a reason/context to the current value of the status/state. It has the relatedEntity and relatedParty in relation to the status/state value.
place	A RelatedPlaceRefOrValue. A list of places (Place [*]). Used to define a place useful for the service (for example a geographical place where the service is installed).
relatedEntity	A RelatedEntityRefOrValue. A list of related entities in relationship with this service.
relatedParty	A RelatedPartyRefOrPartyRoleRef. A list of related party references (RelatedParty [*]). A related party defines party or party role linked to a specific entity.
serviceCharacteristic	A Characteristic. A list of characteristics that characterize this service (ServiceCharacteristic [*]). Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
serviceDate	A String. Date when the service was created (whatever its status).
serviceOrderItem	A RelatedServiceOrderItem. A list of service order items related to this service.
serviceRelationship	A ServiceRelationship. A list of service relationships (ServiceRelationship [*]). Describes links with other service(s) in the inventory.

serviceSpecification	A ServiceSpecificationRef. Service specification reference: ServiceSpecification(s) required to realize a ProductSpecification.
serviceType	A String. Business type of the service.
startDate	A DateTime. Date when the service starts.
startMode	A String. This attribute is an enumerated integer that indicates how the Service is started, such as: 0: Unknown; 1: Automatically by the managed environment; 2: Automatically by the owning device; 3: Manually by the Provider of the Service; 4: Manually by a Customer of the Provider; 5: Any of the above.
state	A ServiceStateType. Valid values for the lifecycle state of the service. ENUMERATED with values: * created * feasibilityChecked * designed * reserved * inactive * pendingActive * active * suspended * pendingTerminate * terminated
supportingResource	A ResourceRef. A list of supporting resources (SupportingResource [*]).Note: only Service of type RFS can be associated with Resources.
supportingService	A ServiceRefOrValue. A list of supporting services (SupportingService [*]). A collection of services that support this service (bundling, link CFS to RFS).
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Feature sub-resource fields

featureCharacteristic	A Characteristic. This is a list of Characteristics for a particular feature. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
featureRelationship	A FeatureRelationship. Collection of feature relationships.
id	A String. Unique identifier.
isBundle	A Boolean. True if this is a feature group. Default is false.
isEnabled	A Boolean. True if this feature is enabled. Default is true.
name	A String. This is the name for the feature.
policyConstraint	A PolicyRef. Reference to manage a list of feature specification policy constraints.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	Feature can be instantiated as * GeographicSiteFeature
attachment	This property is present in subclasses
featureCategory	This property is present in subclasses
note	This property is present in subclasses
relatedParty	This property is present in subclasses
validFor	This property is present in subclasses

Characteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.

@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
value	This property is present in subclasses

CharacteristicRelationship sub-resource fields

id	A String. Unique identifier of the characteristic.
relationshipType	A String. The type of relationship.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BooleanArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Boolean. A characteristic which value is an array of Boolean(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BooleanCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Boolean. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FloatCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Float. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An Integer. A characteristic which value is an array of Integer(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An Integer. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number. A characteristic which value is an array of Number(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An object. Collection of characteristic values.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An object. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A String. Collection of string characteristics.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A String. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FeatureRelationship sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
relationshipType	A String. This is the type of the feature relationship. ENUMERATED with values: * excluded * includes * may include * requires
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PolicyRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.

@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Attachment sub-resource fields

attachmentType	A String. A business characterization of the purpose of the attachment, for example logo, instructionManual, contractCopy.
content	A Base64. The actual contents of the attachment object, if embedded, encoded as base64.
description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
mimeType	A String. A technical characterization of the attachment content format using IETF Mime Types.
name	A String. The name of the attachment.
size	A Quantity. An amount in a given unit.
url	A String. Uniform Resource Locator, is a web page address (a subset of URI).
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

AttachmentRef sub-resource fields

description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
url	A String. Link to the attachment media/content.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

@type	A String. When sub-classing, this defines the sub-class Extensible name.
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Note sub-resource fields

author	A String. Author of the note.
date	A DateTime. Date of the note.
id	A String. Identifier of the note within its containing entity.
text	A String. Text of the note.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CalendarPeriod sub-resource fields

day	A String. Day where the calendar status applies (e.g.: monday, mon-to-fri, weekdays, weekend, all week, ...).
hourPeriod	A HourPeriod. Collection of hour intervals.
status	A String. Indication of the availability of the calendar period (e.g.: available, booked, etc.).
timeZone	A String. Indication of the timezone applicable to the calendar information (e.g.: Paris, GMT+1).
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

HourPeriod sub-resource fields

endHour	A String. The time when the status ends applying.
startHour	A String. The time when the status starts applying.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicSiteFeature sub-resource fields

attachment	An AttachmentRefOrValue. Attachments that may be of relevance to this specification, such as picture, document, media.
featureCategory	A String. Collection of feature category.
featureCharacteristic	A Characteristic. This is a list of Characteristics for a particular feature. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
featureRelationship	A FeatureRelationship. Collection of feature relationships.
id	A String. Unique identifier.
isBundle	A Boolean. True if this is a feature group. Default is false.
isEnabled	A Boolean. True if this feature is enabled. Default is true.
name	A String. This is the name for the feature.
note	A Note. Collection of notes.
policyConstraint	A PolicyRef. Reference to manage a list of feature specification policy constraints.
relatedParty	A RelatedPartyRefOrPartyRoleRef. The associated party or partyRole.
validFor	A CalendarPeriod. Collection of validity information.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Intent sub-resource fields

attachment	An AttachmentRefOrValue. Attachments that may be of relevance to this intent, such as picture, document, media.
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characteristic	A Characteristic. Collection of characteristics. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
context	A String. A string used to give a context to the intent.
creationDate	A DateTime. Date and time of the creation of this REST resource.
description	A String. The description of the intent.
expression	An IntentExpression. An Intent expression. IntentExpression can be instantiated as * JsonLdExpression * RdfXmlExpression * TurtleExpression
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intentRelationship	An EntityRelationship. A list of intents related to this intent.
intentSpecification	An EntityRef. Entity reference schema to be use for all entityRef class.
isBundle	A Boolean. IsBundle determines whether an intent represents a single intent (false), or a bundle of intents(true).
lastUpdate	A DateTime. Date and time of the last update of this REST resource.
lifecycleStatus	A String. Used to indicate the current lifecycle status of this intent.
name	A String. The name of the intent.
priority	A String. Can be used by intent owner to prioritize intents in an intent management system.
relatedParty	A RelatedPartyRefOrPartyRoleRef. Collection of related parties.
statusChangeDate	A DateTime. A date time(DateTime). The date that the entity status changed to the current one.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
version	A String. A field that identifies the specific version of an instance of an intent.

@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntentExpression sub-resource fields

iri	A String. Internationalized Resource Identifier of the intent Expression.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	IntentExpression can be instantiated as * JsonLdExpression * RdfXmlExpression * TurtleExpression
expressionValue	This property is present in subclasses

JsonLdExpressionValue sub-resource fields

@base	A Uri. Used to set the base IRI against which relative IRIs are resolved.
@container	A String. Used to set the default container type for a term. ENUMERATED with values: * @language * @list * @index * @set
@context	An object. Used to define the short-hand names that are used throughout a JSON-LD document.
@graph	A common. Used to express a graph. common can be instantiated as * JsonLdExpressionValue
@id	A Uri. Used to uniquely identify things that are being described in the document with IRIs or blank node identifiers.
@language	A String. Used to specify the language for a particular string value or the default language of a JSON-LD document.
@list	A {}. Used to express an ordered set of data.
@reverse	A String. Used to express reverse properties.
@set	A {}. Used to express an unordered set of data and to ensure that values are always represented as arrays.

@type	A String. Used to set the data type of a node or typed value.
@value	A String. Used to specify the data that is associated with a particular property in the graph.
@vocab	A Uri. Used to expand properties and values in @type with a common prefix IRI.

common sub-resource fields

@base	A Uri. Used to set the base IRI against which relative IRIs are resolved.
@container	A String. Used to set the default container type for a term. ENUMERATED with values: * @language * @list * @index * @set
@id	A Uri. Used to uniquely identify things that are being described in the document with IRIs or blank node identifiers.
@language	A String. Used to specify the language for a particular string value or the default language of a JSON-LD document.
@list	A {}. Used to express an ordered set of data.
@reverse	A String. Used to express reverse properties.
@set	A {}. Used to express an unordered set of data and to ensure that values are always represented as arrays.
@type	A String. Used to set the data type of a node or typed value.
@value	A String. Used to specify the data that is associated with a particular property in the graph.
@vocab	A Uri. Used to expand properties and values in @type with a common prefix IRI.
	common can be instantiated as * JsonLdExpressionValue

common sub-resource fields

See [\[common\]](#) for details.

context sub-resource fields

@context	An object. Used to define the short-hand names that are used throughout a JSON-LD document.
	context can be instantiated as * JsonLdExpressionValue

graph sub-resource fields

@graph	A common. Used to express a graph. common can be instantiated as * JsonLdExpressionValue
	graph can be instantiated as * JsonLdExpressionValue

JsonLdExpression sub-resource fields

expressionValue	A JsonLdExpressionValue. Json LD expression value schema.
iri	A String. Internationalized Resource Identifier of the intent Expression.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RdfXmlExpression sub-resource fields

expressionValue	A String. RDF XML Expression value encoded object.
iri	A String. Internationalized Resource Identifier of the intent Expression.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

TurtleExpression sub-resource fields

expressionValue	A String. Turtle Expression value is the ontology-encoded form of the Intent as Turtle RDF as defined in https://www.w3.org/TR/turtle/ .
iri	A String. Internationalized Resource Identifier of the intent Expression.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

EntityRelationship sub-resource fields

associationSpec	An EntityRef. Entity reference schema to be use for all entityRef class.
href	A String.
id	A String.
name	A String.
relationshipType	A String. Type of relationship such as migration, substitution, dependency, exclusivity.
role	A String. The association role for this entity.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String.

EntityRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

EntityRef sub-resource fields

See [\[EntityRef\]](#) for details.

IntentRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.

@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Note sub-resource fields

See [\[Note\]](#) for details.

ContextUpdate sub-resource fields

id	A String. Identifier of the context update.
lastUpdate	A DateTime. The date and time the status/state last changed.
reason	A String. The reason/context for the current value of the status/state.
relatedEntity	An EntityRef. A list of entities that represent observations, anomalies or managing entities etc associated to the current status/state value.
relatedParty	A RelatedPartyRefOrPartyRoleRef. List of parties associated to the current status/state value.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPlaceRefOrValue sub-resource fields

place	A PlaceRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Place entity and not the PlaceRefOrValue class itself.
role	A String.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeoJsonLineString sub-resource fields

bbox	A bbox. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
geoJson	A LineString. GeoJSON: A collection of Points forming a connected line.

href	A String. Hyperlink reference.
id	A String. Unique identifier.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

LineString sub-resource fields

coordinates	A position. GeoJSON: An array of two or more positions.
type	A String. ENUMERATED with values: * LineString

GeoJsonMultiLineString sub-resource fields

bbox	A bbox. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
geoJson	A MultiLineString. GeoJSON: A collection of distinct LineStrings.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

MultiLineString sub-resource fields

coordinates	A position. GeoJSON: An array of two or more positions.
type	A String. ENUMERATED with values: * MultiLineString

GeoJsonMultiPoint sub-resource fields

bbox	A bbox. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
geoJson	A MultiPoint. GeoJSON: A collection of Points.
href	A String. Hyperlink reference.
id	A String. Unique identifier.

@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

MultiPoint sub-resource fields

coordinates	A position. GeoJSON: An array of positions.
type	A String. ENUMERATED with values: * MultiPoint

GeoJsonMultiPolygon sub-resource fields

bbox	A bbox. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
geoJson	A MultiPolygon. GeoJSON: A collection of Polygons.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

MultiPolygon sub-resource fields

coordinates	A polygon. GeoJSON: An array of polygons.
type	A String. ENUMERATED with values: * MultiPolygon

GeoJsonPoint sub-resource fields

bbox	A bbox. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
geoJson	A Point. GeoJSON: A single position.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

@type	A String. When sub-classing, this defines the sub-class Extensible name.
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Point sub-resource fields

coordinates	A position. GeoJSON: A single position.
type	A String. ENUMERATED with values: * Point

GeoJsonPolygon sub-resource fields

bbox	A bbox. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
geoJson	A Polygon. GeoJSON: An array of linear rings.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Polygon sub-resource fields

coordinates	A linearRing. GeoJSON: An array of linear rings.
type	A String. ENUMERATED with values: * Polygon

GeographicAddress sub-resource fields

city	A String. City that the address is in.
country	A String. Country that the address is in.
countryCode	A StandardIdentifier. Country codes are short alphabetic or numeric geographical codes (geocodes) developed to represent countries and dependent areas, for use in data processing and communications. Several different systems have been developed to do this. The term country code frequently refers to ISO 3166-1 alpha-2 or international dialing codes, the E.164 country calling codes.
externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
geographicAddressRelationship	A GeographicAddressRelationship. A list of geographic addresses relationships. Represents a relationship between geographic addresses.

geographicAddressType	A String. Classification of the address, e.g., residential, industrial.
geographicLocation	A GeographicLocationRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the GeographicLocation entity and not the GeographicLocationRefOrValue class itself.
geographicSite	A GeographicSiteRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the GeographicSite entity and not the GeographicSiteRefOrValue class itself.
geographicSubAddress	A GeographicSubAddress. Collection of sub addresses.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
locality	A String. An area of defined or undefined boundaries within a local authority or other legislatively defined area, usually rural or semi rural in nature. [ANZLIC-STREET], or a suburb, a bounded locality within a city, town or shire principally of urban character [ANZLICSTREET].
postcode	A String. Descriptor for a postal delivery area, used to speed and simplify the delivery of mail (also know as zipcode).
stateOrProvince	A String. The State or Province that the address is in.
streetName	A String. Name of the street or other street type.
streetNr	A String. Number identifying a specific property on a public street. It may be combined with streetNrLast for ranged addresses.
streetNrLast	A String. Last number in a range of street numbers allocated to a property.
streetNrLastSuffix	A String. Last street number suffix for a ranged address.
streetNrSuffix	A String. The first street number suffix.
streetSuffix	A String. A modifier denoting a relative direction.
streetType	A String. Alley, avenue, boulevard, brae, crescent, drive, highway, lane, terrace, parade, place, tarn, way, wharf.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StandardIdentifier sub-resource fields

format	A String. Standard/Regulatory definition identifier. e.g., ISO 3166-1.
href	A String. Hyperlink reference.
id	A String. Unique identifier.

value	A String. The value of the resource in the corresponding standard.e.g., a country code value.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicAddressRelationship sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
relationshipType	A String. Type of relationship between the geographic addresses. This attribute specifies the nature of the relationship between the related addresses. It provides context for understanding how the addresses are related to each other.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicLocation sub-resource fields

bbox	A bbox. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	GeographicLocation can be instantiated as * GeoJsonLineString * GeoJsonMultiLineString * GeoJsonMultiPoint * GeoJsonMultiPolygon * GeoJsonPoint * GeoJsonPolygon

geoJson	This property is present in subclasses
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GeographicLocationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicSite sub-resource fields

calendar	A CalendarPeriod. Collection of calendar items.
code	A String. A code that may be used for some addressing schemes eg: [ANSI T1.253-1999].
contactMedium	A ContactMedium. Collection of contact information. ContactMedium can be instantiated as * EmailContactMedium * FaxContactMedium * GeographicAddressContactMedium * PhoneContactMedium * SocialContactMedium * WebFormContactMedium
creationDate	A DateTime. Date and time when the GeographicSite was created.
description	A String. Text describing additional information regarding the site.
externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
place	A PlaceRefOrValue. Collection of place objects.
relatedParty	A RelatedPartyRefOrPartyRoleRef. Related parties collection.
siteCategory	A String. Site classification/category.
siteFeature	A GeographicSiteFeature. Collection of site features.
siteRelationship	A GeographicSiteRelationship. Collection of site siteRelationships.
status	A String. The condition of the GeographicSite, such as planned, underConstruction, cancelled, active, inactive, former.

@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ContactMedium sub-resource fields

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	ContactMedium can be instantiated as * EmailContactMedium * FaxContactMedium * GeographicAddressContactMedium * PhoneContactMedium * SocialContactMedium * WebFormContactMedium
city	This property is present in subclasses
country	This property is present in subclasses
emailAddress	This property is present in subclasses
faxNumber	This property is present in subclasses
geographicAddress	This property is present in subclasses
phoneNumber	This property is present in subclasses
postCode	This property is present in subclasses
socialNetworkId	This property is present in subclasses
stateOrProvince	This property is present in subclasses
street1	This property is present in subclasses
street2	This property is present in subclasses
url	This property is present in subclasses

GeographicAddressRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

EmailContactMedium sub-resource fields

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
emailAddress	A String. Full email address in standard format.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FaxContactMedium sub-resource fields

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
faxNumber	A String. The fax number of the contact.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicAddressContactMedium sub-resource fields

city	A String. The city.
contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
country	A String. The country.
geographicAddress	A GeographicAddressRef. Reference to a Geographic Address.
id	A String. Identifier for this contact medium.
postCode	A String. Postcode.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
stateOrProvince	A String. State or province.
street1	A String. Describes the street.
street2	A String. Complementary street description.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PhoneContactMedium sub-resource fields

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
id	A String. Identifier for this contact medium.
phoneNumber	A String. The phone number of the contact.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

SocialContactMedium sub-resource fields

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
socialNetworkId	A String. Identifier as a member of a social network.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

WebFormContactMedium sub-resource fields

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
url	A String. Url of the web form.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicSiteRelationship sub-resource fields

href	A String. Reference of the related geographic site.
id	A String. Unique identifier of the related site entity within the server.
relationshipType	A String. Type of relationship.
role	A String. Role of the related site in the relationship.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

@type	A String. When sub-classing, this defines the sub-class Extensible name.
-------	--

GeographicSiteRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicSubAddress sub-resource fields

buildingName	A String. Allows for buildings that have well-known names.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
levelNumber	A String. Used where a level type may be repeated e.g. BASEMENT 1, BASEMENT 2.
levelType	A String. Describes level types within a building.
name	A String. Name of the subAddress to identify it with a meaningful identification.
privateStreetName	A String. Private streets internal to a property (e.g. a university) may have internal names that are not recorded by the land title office.
privateStreetNumber	A String. Private streets numbers internal to a private street.
subAddressType	A String. Type of subAddress : it can be a subunit or a private street.
subUnit	A GeographicSubAddressUnit. Representation of a SubUnit. It is used for describing subunit within a subAddress e.g. BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicSubAddressUnit sub-resource fields

subUnitNumber	A String. The discriminator used for the subunit, often just a simple number but may also be a range.
subUnitType	A String. The type of subunit e.g.BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF, RACK.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicLocation sub-resource fields

See [\[GeographicLocation\]](#) for details.

GeographicSite sub-resource fields

See [\[GeographicSite\]](#) for details.

PlaceRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedEntityRefOrValue sub-resource fields

entity	An EntityRefOrValue.
role	A String.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceRef sub-resource fields

href	A String. Hyperlink reference.
------	--------------------------------

id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ServiceRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ServiceSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Service specification version.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Characteristic sub-resource fields

See [\[Characteristic\]](#) for details.

BooleanArrayCharacteristic sub-resource fields

See [\[BooleanArrayCharacteristic\]](#) for details.

BooleanCharacteristic sub-resource fields

See [\[BooleanCharacteristic\]](#) for details.

FloatCharacteristic sub-resource fields

See [\[FloatCharacteristic\]](#) for details.

IntegerArrayCharacteristic sub-resource fields

See [\[IntegerArrayCharacteristic\]](#) for details.

IntegerCharacteristic sub-resource fields

See [\[IntegerCharacteristic\]](#) for details.

NumberArrayCharacteristic sub-resource fields

See [\[NumberArrayCharacteristic\]](#) for details.

NumberCharacteristic sub-resource fields

See [\[NumberCharacteristic\]](#) for details.

ObjectArrayCharacteristic sub-resource fields

See [\[ObjectArrayCharacteristic\]](#) for details.

ObjectCharacteristic sub-resource fields

See [\[ObjectCharacteristic\]](#) for details.

StringArrayCharacteristic sub-resource fields

See [\[StringArrayCharacteristic\]](#) for details.

StringCharacteristic sub-resource fields

See [\[StringCharacteristic\]](#) for details.

RelatedServiceOrderItem sub-resource fields

itemAction	An OrderItemActionType. Action to be performed on the product. ENUMERATED with values: * add * modify * delete * noChange
itemId	A String. Identifier of the order item where the service was managed.
role	A String. Role of the service order item for this service.

serviceOrderHref	A String. Reference of the related entity.
serviceOrderId	A String. Unique identifier of a related entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ServiceRelationship sub-resource fields

relationshipType	A String.
service	A ServiceRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Service entity and not the ServiceRefOrValue class itself.
serviceRelationshipCharacteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic... Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ServiceSpecificationRef sub-resource fields

See [\[ServiceSpecificationRef\]](#) for details.

ResourceRef sub-resource fields

See [\[ResourceRef\]](#) for details.

ServiceRef sub-resource fields

See [\[ServiceRef\]](#) for details.

AlternateServiceProposal sub-resource fields

alternateService	A ServiceRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Service entity and not the ServiceRefOrValue class itself.
alternateServiceAvailabilityDate	A DateTime. Alternate availability date in case seller is not able to meet requested expected availability date for the service.
id	A String. Identifier of a alternate service proposal.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

EligibilityResultReason sub-resource fields

code	A String. Result reason code.
label	A String. Human readable result reason label.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ServiceQualificationItemRelationship sub-resource fields

id	A String. Id of the serviceQualificationItem (from the same serviceQualificationItem).
relationshipType	A String. Type of relationship (ex: reliesOn, connectedTo, etc...).
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CheckServiceQualificationItem sub-resource fields

alternateServiceProposal	An AlternateServiceProposal. Alternate service proposal is used when the requested service is not available with characteristic and date asked for. An alternate proposal could be a distinct service Spec close to requested one or same as requested but with a different availability date.
appointmentRequired	A Boolean. Indicates whether an appointment is required or not.
category	A ServiceCategoryRef. The (service) category resource is used to group service candidates in logical containers. Categories can contain other categories.
eligibilityResultReason	An EligibilityResultReason. A list of eligibility result reasons (EligibilityResultReason [*]). Reason for eligibility result, whatever the result is (availability or unavailability).
expectedActivationDate	A DateTime. The date when the service is expected to be activated.
expectedServiceAvailabilityDate	A DateTime. Date when the requester looks for service availability.
expirationDate	A DateTime. Date when the qualification item response expires.
id	A String. Id of the Service Qualification Item.
qualificationItemRelationship	A ServiceQualificationItemRelationship. A list of qualification item relationships used to describe relationship between serviceQualification item from the same serviceQualification.
qualificationResult	A String. Qualification result for serviceQualification item. It could be: - qualified (request service are available), - unqualified (requested not available and not alternate available), - alternate (requested not available but proposal available).
service	A ServiceRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Service entity and not the ServiceRefOrValue class itself.
state	A String. State of the CheckServiceQualificationItem: described in the state machine diagram.
terminationError	A TerminationError. If qualificationItem has not been done properly this lists the error(s) that caused termination of the qualification.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ServiceQualificationItem sub-resource fields

See [\[ServiceQualificationItem\]](#) for details.

Json representation sample(s)

We provide below a JSON representation as example of the 'QueryServiceQualification'

resource object.

```
{
  "id": "54",
  "href": "https://mycsp.com:8080/tmf-api/serviceQualificationManagement/v5/queryServiceQualification/54",
  "description": "Query Service Qualification Illustration - 2 serviceSpec available at this place for this category",
  "queryServiceQualificationDate": "2023-04-13T13:51:41.218Z",
  "effectiveQualificationDate": "2023-04-13T13:51:41.218Z",
  "estimatedResponseDate": "2023-04-13T14:00:00.218Z",
  "expectedQualificationDate": "2023-04-13T13:51:41.218Z",
  "expirationDate": "2023-04-15T13:51:41.218Z",
  "instantSyncQualification": true,
  "state": "done",
  "externalIdentifier": [
    {
      "id": "CSQ1",
      "@type": "ExternalIdentifier"
    }
  ],
  "relatedParty": [
    {
      "role": "Buyer contact",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "456",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
        "name": "Jean Pontus",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "searchCriteria": {
    "id": "1",
    "category": {
      "id": "1",
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v5/category/1",
      "name": "Fiber Access",
      "@type": "ServiceCategoryRef",
      "@referredType": "ServiceCategory"
    },
    "service": {
      "place": [
        {
          "@type": "RelatedPlaceRefOrValue",
          "role": "Installation place",
          "place": {
            "@type": "PlaceRef",
            "@referredType": "GeographicSite",
            "id": "11",
            "href": "http://mycsp.com:8080/tmf-api/geographicSiteManagement/v5/geographicSite/11",
            "name": "Jean Pontus home"
          }
        }
      ]
    },
    "@type": "Service"
  },
}
```

```

    "@type": "ServiceQualificationItem"
  },
  "serviceQualificationItem": [
    {
      "id": "1",
      "expirationDate": "2023-04-24T19:20:39.239Z",
      "service": {
        "serviceSpecification": {
          "id": "78",
          "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v5/serviceSpecification/78",
          "name": "X-Fiber Service",
          "@type": "ServiceSpecificationRef",
          "@referredType": "ServiceSpecification"
        },
        "@type": "Service"
      },
      "@type": "ServiceQualificationItem"
    },
    {
      "id": "2",
      "expirationDate": "2023-04-24T19:20:39.239Z",
      "service": {
        "serviceSpecification": {
          "id": "87",
          "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v5/serviceSpecification/87",
          "name": "Standard Fiber Service",
          "@type": "ServiceSpecificationRef",
          "@referredType": "ServiceSpecification"
        },
        "@type": "Service"
      },
      "@type": "ServiceQualificationItem"
    }
  ],
  "@type": "QueryServiceQualification"
}

```

CheckServiceQualification resource

CheckServiceQualification is used to perform a technical eligibility on service configuration(s). It allows to retrieve services that are technically available in the context of the interaction (place, party, service characteristics, ...).

Resource model

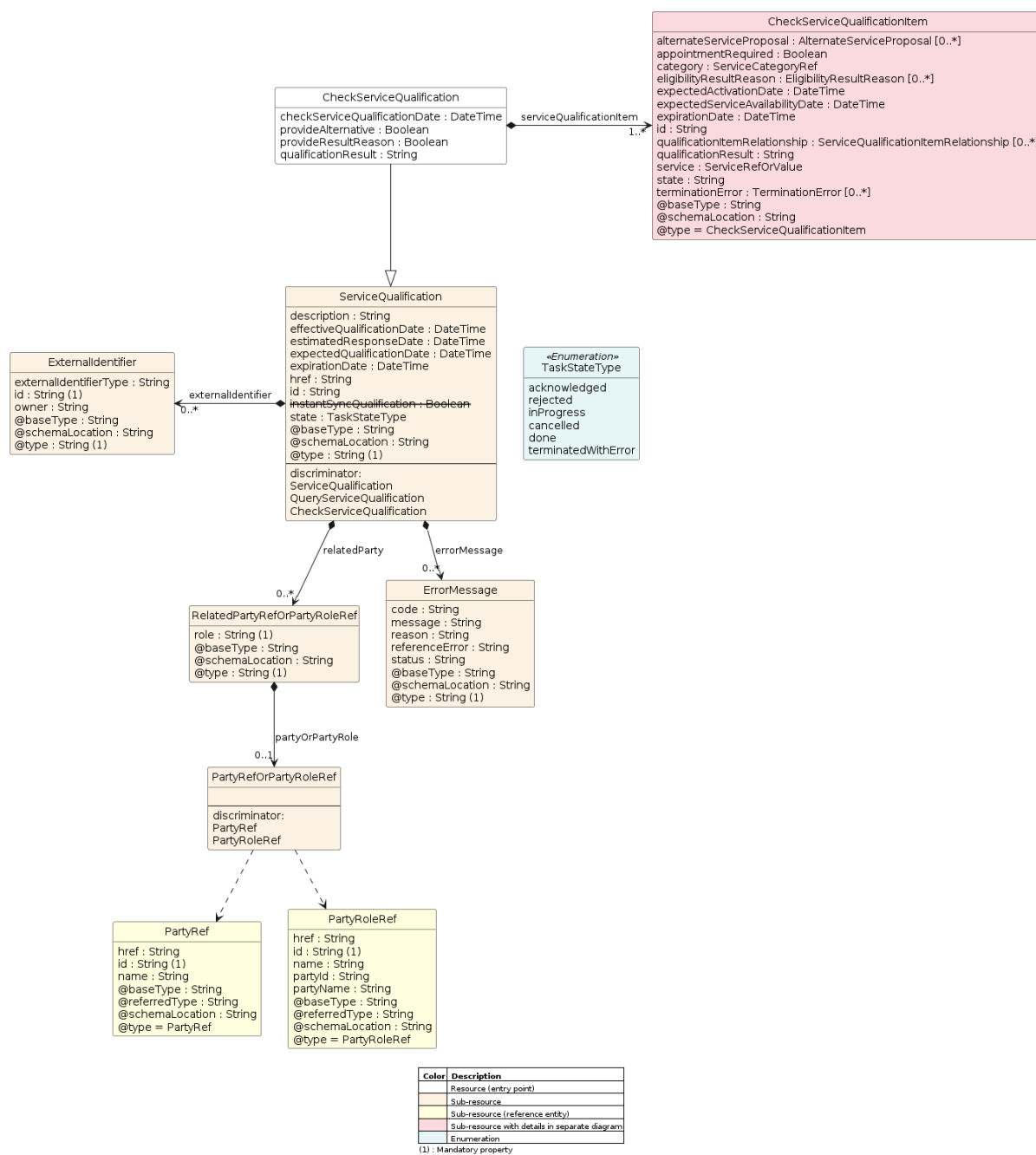


Figure 18 - CheckServiceQualification

See [Figure 2](#) for RelatedEntityRefOrValue diagrams.

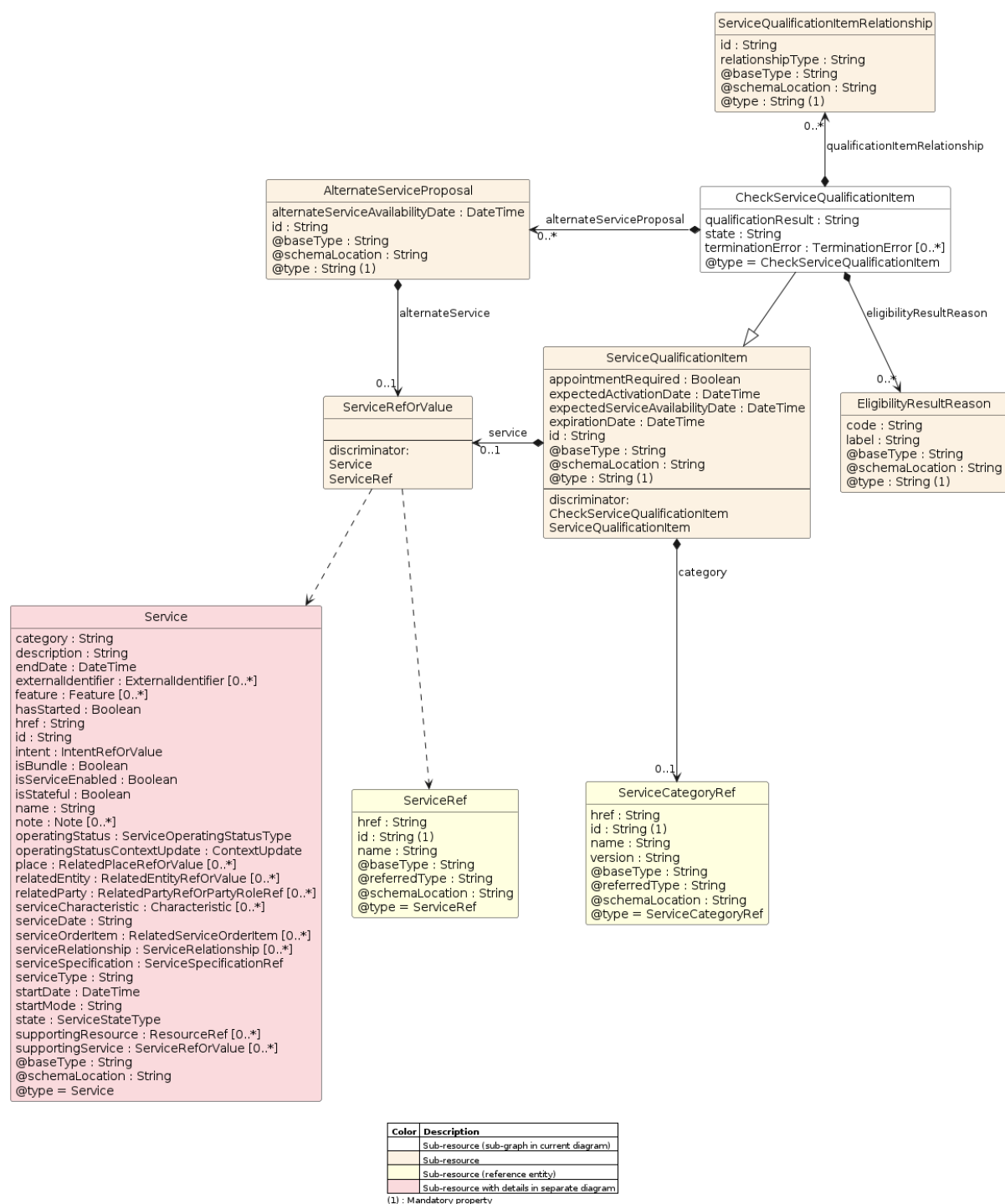


Figure 19 - CheckServiceQualificationItem

See [Figure 4](#) for RelatedPlaceRefOrValue diagrams.

See [Figure 5](#) for IntentExpression diagrams.

See [Figure 6](#) for Feature diagrams.

See [Figure 7](#) for Intent diagrams.

See [Figure 8](#) for EntityRefOrValue diagrams.

See [Figure 9](#) for ContactMedium diagrams.

See [Figure 10](#) for Characteristic diagrams.

See [Figure 11](#) for GeographicAddress diagrams.

See [Figure 12](#) for GeographicSite diagrams.

See [Figure 13](#) for Service diagrams.

See [Figure 14](#) for GeographicLocation diagrams.

See [Figure 15](#) for PlaceRefOrValue diagrams.

See [Figure 16](#) for Place diagrams.

See [Figure 17](#) for GeographicLocationRefOrValue diagrams.

Field descriptions

CheckServiceQualification fields

checkServiceQualificationDate	A DateTime. Date when the serviceQualification was submitted.
description	A String. Description of the serviceQualification.
effectiveQualificationDate	A DateTime. Effective date to serviceQualification completion.
errorMessage	An ErrorMessage. A list of error message contains the error that caused the termination of ServiceQualification task.
estimatedResponseDate	A DateTime. A Date by when the provider estimates supplying a response to the qualification request.
expectedQualificationDate	A DateTime. A date (DateTime). Deadline date when the requester expected a qualification answer.
expirationDate	A DateTime. Date when the qualification response expires.
externalIdentifier	An ExternalIdentifier. A list of external identifiers associated with this resource.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
instantSyncQualification	DEPRECATED A Boolean. A previous indicator which when the value is "true" means that requester expects to get qualification result immediately in the response. If the indicator is true (or by default now without it) then the response code of 200 indicates the operation is successful otherwise a task is created with a response 201. For asynchronous, this attribute is now replaced by the header request "Expect: 202-Accepted".
provideAlternative	A Boolean. When the value is TRUE means that alternative solutions should be provided.

provideResultReason	A Boolean. When the value is TRUE means that result reason are expected for available or not of product offering.
qualificationResult	A String. Qualification result for this serviceQualification. It could be: qualified (all qualification item are qualified), alternate (At least one item alternate and no item with unqualified), unqualified (At least one item unqualified).
relatedParty	A RelatedPartyRefOrPartyRoleRef. A list of related party references, defines party or party role linked to this request.
serviceQualificationItem	A CheckServiceQualificationItem. A list of service qualification items.
state	A TaskStateType. Possible values for the state of a task. ENUMERATED with values: * acknowledged * rejected * inProgress * cancelled * done * terminatedWithError
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ErrorMessage sub-resource fields

See [\[ErrorMessage\]](#) for details.

ExternalIdentifier sub-resource fields

See [\[ExternalIdentifier\]](#) for details.

RelatedPartyRefOrPartyRoleRef sub-resource fields

See [\[RelatedPartyRefOrPartyRoleRef\]](#) for details.

PartyRef sub-resource fields

See [\[PartyRef\]](#) for details.

PartyRoleRef sub-resource fields

See [\[PartyRoleRef\]](#) for details.

CheckServiceQualificationItem sub-resource fields

See [\[CheckServiceQualificationItem\]](#) for details.

AlternateServiceProposal sub-resource fields

See [\[AlternateServiceProposal\]](#) for details.

Service sub-resource fields

See [\[Service\]](#) for details.

Feature sub-resource fields

See [\[Feature\]](#) for details.

Characteristic sub-resource fields

See [\[Characteristic\]](#) for details.

CharacteristicRelationship sub-resource fields

See [\[CharacteristicRelationship\]](#) for details.

BooleanArrayCharacteristic sub-resource fields

See [\[BooleanArrayCharacteristic\]](#) for details.

BooleanCharacteristic sub-resource fields

See [\[BooleanCharacteristic\]](#) for details.

FloatCharacteristic sub-resource fields

See [\[FloatCharacteristic\]](#) for details.

IntegerArrayCharacteristic sub-resource fields

See [\[IntegerArrayCharacteristic\]](#) for details.

IntegerCharacteristic sub-resource fields

See [\[IntegerCharacteristic\]](#) for details.

NumberArrayCharacteristic sub-resource fields

See [\[NumberArrayCharacteristic\]](#) for details.

NumberCharacteristic sub-resource fields

See [\[NumberCharacteristic\]](#) for details.

ObjectArrayCharacteristic sub-resource fields

See [\[ObjectArrayCharacteristic\]](#) for details.

ObjectCharacteristic sub-resource fields

See [\[ObjectCharacteristic\]](#) for details.

StringArrayCharacteristic sub-resource fields

See [\[StringArrayCharacteristic\]](#) for details.

StringCharacteristic sub-resource fields

See [\[StringCharacteristic\]](#) for details.

FeatureRelationship sub-resource fields

See [\[FeatureRelationship\]](#) for details.

PolicyRef sub-resource fields

See [\[PolicyRef\]](#) for details.

Attachment sub-resource fields

See [\[Attachment\]](#) for details.

AttachmentRef sub-resource fields

See [\[AttachmentRef\]](#) for details.

Note sub-resource fields

See [\[Note\]](#) for details.

CalendarPeriod sub-resource fields

See [\[CalendarPeriod\]](#) for details.

HourPeriod sub-resource fields

See [\[HourPeriod\]](#) for details.

GeographicSiteFeature sub-resource fields

See [\[GeographicSiteFeature\]](#) for details.

Intent sub-resource fields

See [\[Intent\]](#) for details.

IntentExpression sub-resource fields

See [\[IntentExpression\]](#) for details.

JsonLdExpressionValue sub-resource fields

See [\[JsonLdExpressionValue\]](#) for details.

common sub-resource fields

See [\[common\]](#) for details.

common sub-resource fields

See [\[common\]](#) for details.

context sub-resource fields

See [\[context\]](#) for details.

graph sub-resource fields

See [\[graph\]](#) for details.

JsonLdExpression sub-resource fields

See [\[JsonLdExpression\]](#) for details.

RdfXmlExpression sub-resource fields

See [\[RdfXmlExpression\]](#) for details.

TurtleExpression sub-resource fields

See [\[TurtleExpression\]](#) for details.

EntityRelationship sub-resource fields

See [\[EntityRelationship\]](#) for details.

EntityRef sub-resource fields

See [\[EntityRef\]](#) for details.

EntityRef sub-resource fields

See [\[EntityRef\]](#) for details.

IntentRef sub-resource fields

See [\[IntentRef\]](#) for details.

Note sub-resource fields

See [\[Note\]](#) for details.

ContextUpdate sub-resource fields

See [\[ContextUpdate\]](#) for details.

RelatedPlaceRefOrValue sub-resource fields

See [\[RelatedPlaceRefOrValue\]](#) for details.

GeoJsonLineString sub-resource fields

See [\[GeoJsonLineString\]](#) for details.

LineString sub-resource fields

See [\[LineString\]](#) for details.

GeoJsonMultiLineString sub-resource fields

See [\[GeoJsonMultiLineString\]](#) for details.

MultiLineString sub-resource fields

See [\[MultiLineString\]](#) for details.

GeoJsonMultiPoint sub-resource fields

See [\[GeoJsonMultiPoint\]](#) for details.

MultiPoint sub-resource fields

See [\[MultiPoint\]](#) for details.

GeoJsonMultiPolygon sub-resource fields

See [\[GeoJsonMultiPolygon\]](#) for details.

MultiPolygon sub-resource fields

See [\[MultiPolygon\]](#) for details.

GeoJsonPoint sub-resource fields

See [\[GeoJsonPoint\]](#) for details.

Point sub-resource fields

See [\[Point\]](#) for details.

GeoJsonPolygon sub-resource fields

See [\[GeoJsonPolygon\]](#) for details.

Polygon sub-resource fields

See [\[Polygon\]](#) for details.

GeographicAddress sub-resource fields

See [\[GeographicAddress\]](#) for details.

StandardIdentifier sub-resource fields

See [\[StandardIdentifier\]](#) for details.

GeographicAddressRelationship sub-resource fields

See [\[GeographicAddressRelationship\]](#) for details.

GeographicLocation sub-resource fields

See [\[GeographicLocation\]](#) for details.

GeographicLocationRef sub-resource fields

See [\[GeographicLocationRef\]](#) for details.

GeographicSite sub-resource fields

See [\[GeographicSite\]](#) for details.

ContactMedium sub-resource fields

See [\[ContactMedium\]](#) for details.

GeographicAddressRef sub-resource fields

See [\[GeographicAddressRef\]](#) for details.

EmailContactMedium sub-resource fields

See [\[EmailContactMedium\]](#) for details.

FaxContactMedium sub-resource fields

See [\[FaxContactMedium\]](#) for details.

GeographicAddressContactMedium sub-resource fields

See [\[GeographicAddressContactMedium\]](#) for details.

PhoneContactMedium sub-resource fields

See [\[PhoneContactMedium\]](#) for details.

SocialContactMedium sub-resource fields

See [\[SocialContactMedium\]](#) for details.

WebFormContactMedium sub-resource fields

See [\[WebFormContactMedium\]](#) for details.

GeographicSiteRelationship sub-resource fields

See [\[GeographicSiteRelationship\]](#) for details.

GeographicSiteRef sub-resource fields

See [\[GeographicSiteRef\]](#) for details.

GeographicSubAddress sub-resource fields

See [\[GeographicSubAddress\]](#) for details.

GeographicSubAddressUnit sub-resource fields

See [\[GeographicSubAddressUnit\]](#) for details.

GeographicLocation sub-resource fields

See [\[GeographicLocation\]](#) for details.

GeographicSite sub-resource fields

See [\[GeographicSite\]](#) for details.

PlaceRef sub-resource fields

See [\[PlaceRef\]](#) for details.

RelatedEntityRefOrValue sub-resource fields

See [\[RelatedEntityRefOrValue\]](#) for details.

ResourceRef sub-resource fields

See [\[ResourceRef\]](#) for details.

ServiceCategoryRef sub-resource fields

See [\[ServiceCategoryRef\]](#) for details.

ServiceRef sub-resource fields

See [\[ServiceRef\]](#) for details.

ServiceSpecificationRef sub-resource fields

See [\[ServiceSpecificationRef\]](#) for details.

Characteristic sub-resource fields

See [\[Characteristic\]](#) for details.

BooleanArrayCharacteristic sub-resource fields

See [\[BooleanArrayCharacteristic\]](#) for details.

BooleanCharacteristic sub-resource fields

See [\[BooleanCharacteristic\]](#) for details.

FloatCharacteristic sub-resource fields

See [\[FloatCharacteristic\]](#) for details.

IntegerArrayCharacteristic sub-resource fields

See [\[IntegerArrayCharacteristic\]](#) for details.

IntegerCharacteristic sub-resource fields

See [\[IntegerCharacteristic\]](#) for details.

NumberArrayCharacteristic sub-resource fields

See [\[NumberArrayCharacteristic\]](#) for details.

NumberCharacteristic sub-resource fields

See [\[NumberCharacteristic\]](#) for details.

ObjectArrayCharacteristic sub-resource fields

See [\[ObjectArrayCharacteristic\]](#) for details.

ObjectCharacteristic sub-resource fields

See [\[ObjectCharacteristic\]](#) for details.

StringArrayCharacteristic sub-resource fields

See [\[StringArrayCharacteristic\]](#) for details.

StringCharacteristic sub-resource fields

See [\[StringCharacteristic\]](#) for details.

RelatedServiceOrderItem sub-resource fields

See [\[RelatedServiceOrderItem\]](#) for details.

ServiceRelationship sub-resource fields

See [\[ServiceRelationship\]](#) for details.

ServiceSpecificationRef sub-resource fields

See [\[ServiceSpecificationRef\]](#) for details.

ResourceRef sub-resource fields

See [\[ResourceRef\]](#) for details.

ServiceRef sub-resource fields

See [\[ServiceRef\]](#) for details.

ServiceCategoryRef sub-resource fields

See [\[ServiceCategoryRef\]](#) for details.

EligibilityResultReason sub-resource fields

See [\[EligibilityResultReason\]](#) for details.

ServiceQualificationItemRelationship sub-resource fields

See [\[ServiceQualificationItemRelationship\]](#) for details.

Service sub-resource fields

See [\[Service\]](#) for details.

ServiceRef sub-resource fields

See [\[ServiceRef\]](#) for details.

ServiceQualificationItem sub-resource fields

See [\[ServiceQualificationItem\]](#) for details.

Json representation sample(s)

We provide below a JSON representation as example of the 'CheckServiceQualification' resource object.

```
{
  "id": "456",
  "href": "https://mycsp.com:8080/tmf-api/serviceQualificationManagement/v5/checkServiceQualification/456",
  "checkServiceQualificationDate": "2023-04-10T19:20:39.239Z",
  "description": "check service qualification illustration",
  "effectiveQualificationDate": "2023-04-11T15:20:39.239Z",
  "estimatedResponseDate": "2023-04-11T15:20:39.239Z",
  "expectedQualificationDate": "2023-04-11T15:20:39.239Z",
  "expirationDate": "2023-04-24T19:20:39.239Z",
  "externalIdentifier": [
    {
      "id": "CSQ1",
      "@type": "ExternalIdentifier"
    }
  ],
  "provideAlternative": true,
  "provideResultReason": true,
  "instantSyncQualification": false,
  "qualificationResult": "alternate",
  "relatedParty": [
    {
      "role": "Buyer contact",
```

```

    "@type": "RelatedPartyRefOrPartyRoleRef",
    "partyOrPartyRole": {
      "id": "456",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
      "name": "Jean Pontus",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  ],
  "serviceQualificationItem": [
    {
      "id": "1",
      "expectedActivationDate": "2023-04-16T19:20:39.239Z",
      "expectedServiceAvailabilityDate": "2023-04-11T19:20:39.239Z",
      "expirationDate": "2023-04-24T19:20:39.239Z",
      "qualificationResult": "alternate",
      "state": "done",
      "alternateServiceProposal": [
        {
          "id": "1",
          "alternateServiceAvailabilityDate": "2023-04-15T19:20:39.239Z",
          "alternateService": {
            "serviceCharacteristic": [
              {
                "id": "1",
                "name": "BroadcastQuality",
                "valueType": "string",
                "value": "HD",
                "@type": "StringCharacteristic"
              }
            ],
            "serviceSpecification": {
              "id": "23",
              "href": "https://mycsp.com:8080/tmf-
api/serviceCatalogManagement/v5/serviceSpecification/23",
              "name": "TVAccess",
              "@type": "ServiceSpecificationRef",
              "@referredType": "ServiceSpecification"
            },
            "@type": "Service"
          },
          "@type": "AlternateServiceProposal"
        }
      ],
      "eligibilityResultReason": [
        {
          "code": "26-001",
          "label": "not enough bandwidth at this place",
          "@type": "EligibilityResultReason"
        }
      ],
      "service": {
        "place": [
          {
            "role": "Installation place",
            "@type": "RelatedPlaceRefOrValue",
            "place": {
              "@type": "PlaceRef",
              "@referredType": "GeographicSite",
              "id": "11",
              "href": "https://mycsp.com:8080/tmf-
api/geographicSiteManagement/v5/geographicSite/11",

```

```

        "name": "Jean Pontus home"
      }
    ],
    "serviceCharacteristic": [
      {
        "id": "1",
        "name": "BroadcastQuality",
        "valueType": "string",
        "value": "4K",
        "@type": "StringCharacteristic"
      }
    ],
    "serviceSpecification": {
      "id": "23",
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v5/serviceSpecification/23",
      "name": "TVAccess",
      "@type": "ServiceSpecificationRef",
      "@referredType": "ServiceSpecification"
    },
    "@type": "Service"
  },
  "@type": "CheckServiceQualificationItem"
}
],
"state": "done",
"@type": "CheckServiceQualification"
}

```

Notification Resource Models

8 notifications are defined for this API.

Notifications related to QueryServiceQualification:

- Create Event
- Delete Event
- State Change Event

Notifications related to CheckServiceQualification:

- Create Event
- Delete Event
- Attribute Value Change Event
- Information Required Event
- State Change Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a subclass of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the

resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

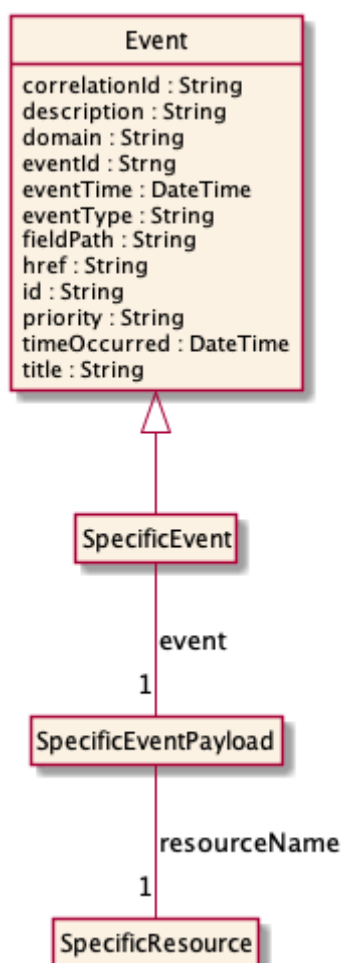


Figure 20 Notification Pattern

QueryServiceQualification

Create Event

Message example for QueryServiceQualificationCreateEvent event

```

Content-Type: application/json
{
  "correlationId": "b122fa39-7242",
  "description": "QueryServiceQualificationCreateEvent illustration",
  "domain": "Commercial",
  "eventId": "4d6e-98cf-a085d0121f61",
  "eventTime": "2022-08-25T12:20:45.491Z",
  "eventType": "QueryServiceQualificationCreateEvent",
  "priority": "5",
  "timeOccurred": "2022-08-25T12:20:44.232Z",
  "title": "QueryServiceQualificationCreateEvent",
  "event": {
    "queryServiceQualification": {
      "id": "54",
      "href": "https://mycsp.com:8080/tmf-
  
```

```

api/serviceQualificationManagement/v5/queryServiceQualification/54",
  "description": "Query Service Qualification Illustration - 2 serviceSpec
available at this place for this category",
  "effectiveQualificationDate": "2020-04-13T13:51:41.218Z",
  "estimatedResponseDate": "2020-04-13T14:00:00.218Z",
  "expectedQualificationDate": "2020-04-13T13:51:41.218Z",
  "expirationDate": "2020-04-15T13:51:41.218Z",
  "externalIdentifier": [
    {
      "id": "CSQ1",
      "@type": "ExternalIdentifier"
    }
  ],
  "instantSyncQualification": true,
  "queryServiceQualificationDate": "2020-04-13T13:51:41.218Z",
  "relatedParty": [
    {
      "role": "Buyer contact",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "456",
        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/456",
        "name": "Jean Pontus",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "searchCriteria": {
    "id": "1",
    "category": {
      "id": "1",
      "href": "https://mycsp.com:8080/tmf-
api/serviceCatalogManagement/v5/category/1",
      "name": "Fiber Access",
      "@type": "ServiceCategoryRef",
      "@referredType": "ServiceCategory"
    },
    "service": {
      "place": [
        {
          "@type": "RelatedPlaceRefOrValue",
          "role": "Installation place",
          "place": {
            "@type": "PlaceRef",
            "@referredType": "GeographicSite",
            "id": "11",
            "href": "http://mycsp.com:8080/tmf-
api/geographicSiteManagement/v5/geographicSite/11",
            "name": "Jean Pontus home"
          }
        }
      ],
      "@type": "Service"
    },
    "@type": "ServiceQualificationItem"
  },
  "serviceQualificationItem": [
    {
      "id": "1",
      "expirationDate": "2023-04-24T19:20:39.239Z",
      "service": {

```

```

        "serviceSpecification": {
            "id": "78",
            "href": "https://mycsp.com:8080/tmf-
api/serviceCatalogManagement/v5/serviceSpecification/78",
            "name": "X-Fiber Service",
            "@type": "ServiceSpecificationRef",
            "@referredType": "ServiceSpecification"
        },
        "@type": "Service"
    },
    "@type": "ServiceQualificationItem"
},
{
    "id": "2",
    "expirationDate": "2023-04-24T19:20:39.239Z",
    "service": {
        "serviceSpecification": {
            "id": "87",
            "href": "https://mycsp.com:8080/tmf-
api/serviceCatalogManagement/v5/serviceSpecification/87",
            "name": "Standard Fiber Service",
            "@type": "ServiceSpecificationRef",
            "@referredType": "ServiceSpecification"
        },
        "@type": "Service"
    },
    "@type": "ServiceQualificationItem"
}
],
"state": "done",
"@type": "QueryServiceQualification"
}
},
"reportingSystem": {
    "id": "768",
    "name": "APP-897",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
},
"source": {
    "id": "276",
    "name": "APP-819",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
},
"@baseType": "Event",
"@type": "QueryServiceQualificationCreateEvent"
}

```

Delete Event

Message example for QueryServiceQualificationDeleteEvent event

```

Content-Type: application/json
{
    "correlationId": "4afc7d04-5251",
    "description": "QueryServiceQualificationDeleteEvent illustration",
    "domain": "Commercial",
    "eventId": "4212-968d-a8fe5f08aa16",
    "eventTime": "2022-08-25T12:20:45.513Z",

```

```

    "eventType": "QueryServiceQualificationDeleteEvent",
    "priority": "2",
    "timeOccurred": "2022-08-25T12:20:36.791Z",
    "title": "QueryServiceQualificationDeleteEvent",
    "event": {
      "queryServiceQualification": {
        "id": "54",
        "href": "http://mycsp.com:8080/tmf-
api/serviceQualificationManagement/v5/queryServiceQualification/54",
        "@type": "QueryServiceQualification"
      }
    },
    "reportingSystem": {
      "id": "768",
      "name": "APP-897",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "source": {
      "id": "276",
      "name": "APP-819",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "QueryServiceQualificationDeleteEvent"
  }

```

State Change Event

Message example for QueryServiceQualificationStateChangeEvent event

```

Content-Type: application/json
{
  "correlationId": "95e2f5d5-e67f",
  "description": "QueryServiceQualificationStateChangeEvent illustration",
  "domain": "Commercial",
  "eventId": "4011-abc1-8a9dfc8ff694",
  "eventTime": "2022-08-25T12:20:45.501Z",
  "eventType": "QueryServiceQualificationStateChangeEvent",
  "priority": "5",
  "timeOccurred": "2022-08-25T12:20:40.538Z",
  "title": "QueryServiceQualificationStateChangeEvent",
  "event": {
    "queryServiceQualification": {
      "href": "https://mycsp.com:8080/tmf-
api/serviceQualificationManagement/v5/queryServiceQualification/54",
      "id": "54",
      "@type": "QueryServiceQualification",
      "state": "done"
    }
  },
  "reportingSystem": {
    "id": "768",
    "name": "APP-897",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "276",

```

```

    "name": "APP-819",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "QueryServiceQualificationStateChangeEvent"
}

```

CheckServiceQualification

Create Event

Message example for CheckServiceQualificationCreateEvent event

```

Content-Type: application/json
{
  "correlationId": "901e0ff8-b717",
  "description": "CheckServiceQualificationCreateEvent illustration",
  "domain": "Commercial",
  "eventId": "439e-8ffd-25b84f13b3df",
  "eventTime": "2023-03-05T12:41:51.931Z",
  "eventType": "CheckServiceQualificationCreateEvent",
  "priority": "2",
  "timeOccurred": "2023-03-05T12:41:49.125Z",
  "title": "CheckServiceQualificationCreateEvent",
  "event": {
    "checkServiceQualification": {
      "id": "44",
      "href": "https://mycsp.com:8080/tmf-api/serviceQualificationManagement/v5/checkServiceQualification/44",
      "checkServiceQualificationDate": "2023-04-10T19:20:39.239Z",
      "description": "check service qualification illustration",
      "effectiveQualificationDate": "2023-04-11T15:20:39.239Z",
      "estimatedResponseDate": "2023-04-11T15:20:39.239Z",
      "expectedQualificationDate": "2023-04-11T15:20:39.239Z",
      "expirationDate": "2023-04-24T19:20:39.239Z",
      "externalIdentifier": [
        {
          "id": "CSQ1",
          "@type": "ExternalIdentifier"
        }
      ],
      "provideAlternative": true,
      "provideResultReason": true,
      "instantSyncQualification": false,
      "qualificationResult": "alternate",
      "relatedParty": [
        {
          "role": "Buyer contact",
          "@type": "RelatedPartyRefOrPartyRoleRef",
          "partyOrPartyRole": {
            "id": "456",
            "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
            "name": "Jean Pontus",
            "@type": "PartyRef",
            "@referredType": "Individual"
          }
        }
      ]
    }
  }
}

```

```

    ],
    "serviceQualificationItem": [
      {
        "id": "1",
        "expectedActivationDate": "2023-04-16T19:20:39.239Z",
        "expectedServiceAvailabilityDate": "2023-04-11T19:20:39.239Z",
        "expirationDate": "2023-04-24T19:20:39.239Z",
        "qualificationResult": "alternate",
        "state": "done",
        "alternateServiceProposal": [
          {
            "id": "1",
            "alternateServiceAvailabilityDate": "2023-04-15T19:20:39.239Z",
            "alternateService": {
              "serviceCharacteristic": [
                {
                  "id": "1",
                  "name": "BroadcastQuality",
                  "valueType": "string",
                  "value": "HD",
                  "@type": "StringCharacteristic"
                }
              ],
              "serviceSpecification": {
                "id": "23",
                "href": "https://mycsp.com:8080/tmf-
api/serviceCatalogManagement/v5/serviceSpecification/23",
                "name": "TVAccess",
                "@type": "ServiceSpecificationRef",
                "@referredType": "ServiceSpecification"
              },
              "@type": "Service"
            },
            "@type": "AlternateServiceProposal"
          }
        ],
        "eligibilityResultReason": [
          {
            "code": "26-001",
            "label": "not enough bandwidth at this place",
            "@type": "EligibilityResultReason"
          }
        ],
        "service": {
          "place": [
            {
              "role": "Installation place",
              "@type": "RelatedPlaceRefOrValue",
              "place": {
                "@type": "PlaceRef",
                "@referredType": "GeographicSite",
                "id": "11",
                "href": "https://mycsp.com:8080/tmf-
api/geographicSiteManagement/v5/geographicSite/11",
                "name": "Jean Pontus home"
              }
            }
          ],
          "serviceCharacteristic": [
            {
              "@type": "StringCharacteristic",
              "id": "1",
              "name": "BroadcastQuality",

```

```

        "valueType": "string",
        "value": "4K"
      }
    ],
    "serviceSpecification": {
      "id": "23",
      "href": "https://mycsp.com:8080/tmf-
api/serviceCatalogManagement/v5/serviceSpecification/23",
      "name": "TVAccess",
      "@type": "ServiceSpecificationRef",
      "@referredType": "ServiceSpecification"
    },
    "@type": "Service"
  },
  "@type": "ServiceQualificationItem"
}
],
"state": "done",
"@type": "CheckServiceQualification"
}
},
"reportingSystem": {
  "id": "954",
  "name": "APP-190",
  "@type": "ReportingResource",
  "@referredType": "LogicalResource"
},
"source": {
  "id": "441",
  "name": "APP-643",
  "@type": "ReportingResource",
  "@referredType": "LogicalResource"
},
"@baseType": "Event",
"@type": "CheckServiceQualificationCreateEvent"
}

```

Delete Event

Message example for CheckServiceQualificationDeleteEvent event

```

Content-Type: application/json
{
  "correlationId": "b3727ad0-1140",
  "description": "CheckServiceQualificationDeleteEvent illustration",
  "domain": "Commercial",
  "eventId": "4c58-840e-790e0657230f",
  "eventTime": "2023-03-05T12:41:51.932Z",
  "eventType": "CheckServiceQualificationDeleteEvent",
  "priority": "4",
  "timeOccurred": "2023-03-05T12:41:46.084Z",
  "title": "CheckServiceQualificationDeleteEvent",
  "event": {
    "checkServiceQualification": {
      "id": "44",
      "href": "https://mycsp.com:8080/tmf-
api/serviceQualificationManagement/v5/checkServiceQualification/44",
      "@type": "CheckServiceQualification"
    }
  }
},

```

```

"reportingSystem": {
  "id": "954",
  "name": "APP-190",
  "@type": "ReportingResource",
  "@referredType": "LogicalResource"
},
"source": {
  "id": "441",
  "name": "APP-643",
  "@type": "ReportingResource",
  "@referredType": "LogicalResource"
},
"@baseType": "Event",
"@type": "CheckServiceQualificationDeleteEvent"
}

```

Attribute Value Change Event

Message example for CheckServiceQualificationAttributeValueChangeEvent event

```

Content-Type: application/json
{
  "correlationId": "e7e44c9a-64f5",
  "description": "CheckServiceQualificationAttributeValueChangeEvent illustration",
  "domain": "Commercial",
  "eventId": "483f-800d-f34613c8d3e2",
  "eventTime": "2023-08-25T12:20:45.443Z",
  "eventType": "CheckServiceQualificationAttributeValueChangeEvent",
  "priority": "1",
  "timeOccurred": "2023-08-25T12:20:42.512Z",
  "title": "CheckServiceQualificationAttributeValueChangeEvent",
  "event": {
    "checkServiceQualification": {
      "href": "https://mycsp.com:8080/tmf-
api/serviceQualificationManagement/v5/checkServiceQualification/44",
      "id": "44",
      "@type": "CheckServiceQualification",
      "relatedParty": [
        {
          "role": "Buyer contact",
          "@type": "RelatedPartyRefOrPartyRoleRef",
          "partyOrPartyRole": {
            "id": "456",
            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/456",
            "name": "Ross",
            "@type": "PartyRef",
            "@referredType": "Individual"
          }
        }
      ]
    }
  },
  "reportingSystem": {
    "id": "768",
    "name": "APP-897",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {

```

```

    "id": "276",
    "name": "APP-819",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "CheckServiceQualificationAttributeValueChangeEvent"
}

```

Information Required Event

Message example for CheckServiceQualificationInformationRequiredEvent event

```

Content-Type: application/json
{
  "correlationId": "d4c8651f-3939",
  "description": "CheckServiceQualificationInformationRequiredEvent illustration",
  "domain": "Commercial",
  "eventId": "45bd-a19f-814d80a49d5e",
  "eventTime": "2023-08-25T12:20:45.475Z",
  "eventType": "CheckServiceQualificationInformationRequiredEvent",
  "priority": "4",
  "timeOccurred": "2023-08-25T12:20:40.050Z",
  "title": "CheckServiceQualificationInformationRequiredEvent",
  "event": {
    "checkServiceQualification": {
      "id": "44",
      "href": "http://mycsp.com:8080/tmf-api/serviceQualificationManagement/v5/checkServiceQualification/44",
      "@type": "CheckServiceQualification"
    },
    "informationRequired": {
      "informationRequiredPath": [
        {
          "op": "add",
          "path": "expectedQualificationDate"
        }
      ]
    }
  },
  "reportingSystem": {
    "id": "768",
    "name": "APP-897",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "276",
    "name": "APP-819",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "CheckServiceQualificationInformationRequiredEvent"
}

```

State Change Event

Message example for CheckServiceQualificationStateChangeEvent event with ?fields=state property only

```
Content-Type: application/json
{
  "correlationId": "a19c7353-eae1",
  "description": "CheckServiceQualificationStateChangeEvent illustration",
  "domain": "Commercial",
  "eventId": "45a4-81bb-85fa6cc903c8",
  "eventTime": "2023-03-05T12:41:51.931Z",
  "eventType": "CheckServiceQualificationStateChangeEvent",
  "priority": "5",
  "timeOccurred": "2023-03-05T12:41:44.375Z",
  "title": "CheckServiceQualificationStateChangeEvent",
  "event": {
    "checkServiceQualification": {
      "href": "https://mycsp.com:8080/tmf-
api/serviceQualificationManagement/v5/checkServiceQualification/44",
      "id": "44",
      "@type": "CheckServiceQualification",
      "state": "done"
    }
  },
  "reportingSystem": {
    "id": "954",
    "name": "APP-190",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "441",
    "name": "APP-643",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "CheckServiceQualificationStateChangeEvent"
}
```

API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on QueryServiceQualification

List or find QueryServiceQualification objects

GET /queryServiceQualification?fields=...

Description

This operation list QueryServiceQualification entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a list of query service qualification request(s). The given criteria is the query service qualification state (done).

Request

```
GET /queryServiceQualification?filter=state=done
Content-Type: application/json
```

Response

```

200
[
  {
    "id": "54",
    "href": "https://mycsp.com:8080/tmf-
api/serviceQualificationManagement/v5/queryServiceQualification/54",
    "description": "Query Service Qualification Illustration - 2 serviceSpec
available at this place for this category",
    "effectiveQualificationDate": "2023-04-13T13:51:41.218Z",
    "estimatedResponseDate": "2023-04-13T14:00:00.218Z",
    "expectedQualificationDate": "2023-04-13T13:51:41.218Z",
    "expirationDate": "2023-04-15T13:51:41.218Z",
    "externalIdentifier": [
      {
        "id": "CSQ1",
        "@type": "ExternalIdentifier"
      }
    ],
    "instantSyncQualification": true,
    "queryServiceQualificationDate": "2023-04-13T13:51:41.218Z",
    "relatedParty": [
      {
        "role": "Buyer contact",
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "id": "456",
          "href": "http://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
          "name": "Jean Pontus",
          "@referredType": "Individual"
        }
      }
    ],
    "searchCriteria": {
      "id": "1",
      "category": {
        "id": "1",
        "href": "http://mycsp.com:8080/tmf-
api/serviceCatalogManagement/v5/category/1",
        "name": "Fiber Access",
        "@type": "ServiceCategoryRef",
        "@referredType": "ServiceCategory"
      },
      "service": {
        "place": [
          {
            "@type": "RelatedPlaceRefOrValue",
            "role": "Installation place",
            "place": {
              "@type": "PlaceRef",
              "@referredType": "GeographicSite",
              "id": "11",
              "href": "http://mycsp.com:8080/tmf-
api/geographicSiteManagement/v5/geographicSite/11",
              "name": "Jean Pontus home"
            }
          }
        ],
        "@type": "Service"
      },
      "@type": "ServiceQualificationItem"
    }
  ],

```

```

    "serviceQualificationItem": [
      {
        "id": "1",
        "service": {
          "serviceSpecification": {
            "id": "78",
            "href": "http://mycsp.com:8080/tmf-
api/serviceCatalogManagement/v5/serviceSpecification/78",
            "name": "X-Fiber Service",
            "@type": "ServiceSpecificationRef",
            "@referredType": "ServiceSpecification"
          },
          "@type": "Service"
        },
        "@type": "ServiceQualificationItem"
      },
      {
        "id": "2",
        "service": {
          "serviceSpecification": {
            "id": "87",
            "href": "http://mycsp.com:8080/tmf-
api/serviceCatalogManagement/v5/serviceSpecification/87",
            "name": "Standard Fiber Service",
            "@type": "ServiceSpecificationRef",
            "@referredType": "ServiceSpecification"
          },
          "@type": "Service"
        },
        "@type": "ServiceQualificationItem"
      }
    ],
    "state": "done",
    "@type": "QueryServiceQualification"
  }
]

```

Here's an example of a request for retrieving a list of query service qualification request(s) with fields selection for id, effectiveQualificationDate & expirationDate

Request

```

GET
/queryServiceQualification?fields=id,effectiveQualificationDate,expirationDate&filter=state=done
Content-Type: application/json

```

Response

```

200
[
  {
    "id": "48",
    "href": "https://mycsp.com:8080/tmf-
api/serviceQualificationManagement/v5/queryServiceQualification/48",
    "effectiveQualificationDate": "2023-04-11T14:23:33.239Z",
    "expirationDate": "2023-05-11T14:23:33.239Z",

```

```

    "@type": "QueryServiceQualification"
  },
  {
    "id": "44",
    "href": "https://mycsp.com:8080/tmf-
api/serviceQualificationManagement/v5/queryServiceQualification/44",
    "effectiveQualificationDate": "2023-04-12T14:23:33.239Z",
    "expirationDate": "2023-05-11T14:23:33.239Z",
    "@type": "QueryServiceQualification"
  }
]

```

Retrieves a QueryServiceQualification by ID

GET /queryServiceQualification/{id}?fields=...

Description

This operation retrieves a QueryServiceQualification entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a query service qualification request retrieval. The given criteria is the query service qualification for id 54.

Request

```

GET /queryServiceQualification/54
Content-Type: application/json

```

Response

```

200
{
  "id": "54",
  "href": "https://mycsp.com:8080/tmf-
api/serviceQualificationManagement/v5/queryServiceQualification/54",
  "description": "Query Service Qualification Illustration - 2 serviceSpec available
at this place for this category",
  "effectiveQualificationDate": "2020-04-13T13:51:41.218Z",
  "estimatedResponseDate": "2023-04-13T14:00:00.218Z",
  "expectedQualificationDate": "2023-04-13T13:51:41.218Z",
  "expirationDate": "2023-04-15T13:51:41.218Z",
  "externalIdentifier": [
    {
      "id": "CSQ1",
      "@type": "ExternalIdentifier"
    }
  ],
  "instantSyncQualification": true,
  "queryServiceQualificationDate": "2023-04-13T13:51:41.218Z",
  "relatedParty": [
    {

```

```

    "role": "Buyer contact",
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "partyOrPartyRole": {
      "@type": "PartyRef",
      "id": "456",
      "href": "http://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
      "name": "Jean Pontus",
      "@referredType": "Individual"
    }
  },
  "searchCriteria": {
    "id": "1",
    "category": {
      "id": "1",
      "href": "http://mycsp.com:8080/tmf-api/serviceCatalogManagement/v5/category/1",
      "name": "Fiber Access",
      "@type": "ServiceCategoryRef",
      "@referredType": "ServiceCategory"
    },
    "service": {
      "place": [
        {
          "@type": "RelatedPlaceRefOrValue",
          "role": "Installation place",
          "place": {
            "@type": "PlaceRef",
            "@referredType": "GeographicSite",
            "id": "11",
            "href": "http://mycsp.com:8080/tmf-api/geographicSiteManagement/v5/geographicSite/11",
            "name": "Jean Pontus home"
          }
        }
      ],
      "@type": "Service"
    },
    "@type": "ServiceQualificationItem"
  },
  "serviceQualificationItem": [
    {
      "id": "1",
      "service": {
        "serviceSpecification": {
          "id": "78",
          "href": "http://mycsp.com:8080/tmf-api/serviceCatalogManagement/v5/serviceSpecification/78",
          "name": "X-Fiber Service",
          "@type": "ServiceSpecificationRef",
          "@referredType": "ServiceSpecification"
        },
        "@type": "Service"
      },
      "@type": "ServiceQualificationItem"
    },
    {
      "id": "2",
      "service": {
        "serviceSpecification": {
          "id": "87",
          "href": "http://mycsp.com:8080/tmf-api/serviceCatalogManagement/v5/serviceSpecification/87",

```

```

        "name": "Standard Fiber Service",
        "@type": "ServiceSpecificationRef",
        "@referredType": "ServiceSpecification"
      },
      "@type": "Service"
    },
    "@type": "ServiceQualificationItem"
  }
],
"state": "done",
"@type": "QueryServiceQualification"
}

```

Initiate a task for QueryServiceQualification

POST /queryServiceQualification?fields=...

Description

This operation creates a QueryServiceQualification entity.

Mandatory Attributes

Mandatory Attributes	Rule
searchCriteria	
@type	M

Usage samples

Here's an example of a request for creating QueryServiceQualification.

Request

```

POST /queryServiceQualification
Content-Type: application/json
{
  "description": "Query Service Qualification POST Illustration ",
  "expectedQualificationDate": "2023-04-13T13:51:41.218Z",
  "instantSyncQualification": false,
  "externalIdentifier": [
    {
      "id": "CSQ1",
      "@type": "ExternalIdentifier"
    }
  ],
  "relatedParty": [
    {
      "role": "Buyer contact",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "id": "456",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
        "name": "Jean Pontus",
        "@referredType": "Individual"
      }
    }
  ]
}

```

```

    }
  },
  "searchCriteria": {
    "id": "1",
    "category": {
      "id": "1",
      "href": "https://mycsp.com:8080/tmf-
api/serviceCatalogManagement/v5/category/1",
      "name": "Fiber Access",
      "@type": "ServiceCategoryRef",
      "@referredType": "ServiceCategory"
    },
    "service": {
      "place": [
        {
          "@type": "RelatedPlaceRefOrValue",
          "role": "Installation place",
          "place": {
            "@type": "PlaceRef",
            "@referredType": "GeographicSite",
            "id": "11",
            "href":
"http://serverlocation:port/geographicSiteManagement/v5/geographicSite/11",
            "name": "Jean Pontus home"
          }
        }
      ],
      "serviceSpecification": {
        "id": "78",
        "href": "https://mycsp.com:8080/tmf-
api/serviceCatalogManagement/v5/serviceSpecification/78",
        "@type": "ServiceSpecificationRef",
        "@referredType": "ServiceSpecification"
      },
      "@type": "Service"
    },
    "@type": "ServiceQualificationItem"
  },
  "@type": "QueryServiceQualification"
}

```

Response

```

201
{
  "id": "54",
  "href": "https://mycsp.com:8080/tmf-
api/serviceQualificationManagement/v5/queryServiceQualification/54",
  "description": "Query Service Qualification Illustration - 2 serviceSpec available
at this place for this category",
  "effectiveQualificationDate": "2023-04-13T13:51:41.218Z",
  "estimatedResponseDate": "2023-04-13T14:00:00.218Z",
  "expectedQualificationDate": "2023-04-13T13:51:41.218Z",
  "expirationDate": "2023-04-15T13:51:41.218Z",
  "externalIdentifier": [
    {
      "id": "CSQ1",
      "@type": "ExternalIdentifier"
    }
  ]
}

```

```

    ],
    "instantSyncQualification": false,
    "queryServiceQualificationDate": "2023-04-13T13:51:41.218Z",
    "relatedParty": [
      {
        "role": "Buyer contact",
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "partyOrPartyRole": {
          "id": "456",
          "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
          "name": "Jean Pontus",
          "@type": "PartyRef",
          "@referredType": "Individual"
        }
      }
    ],
    "searchCriteria": {
      "id": "1",
      "category": {
        "id": "1",
        "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v5/category/1",
        "name": "Fiber Access",
        "@type": "ServiceCategoryRef",
        "@referredType": "ServiceCategory"
      },
      "service": {
        "place": [
          {
            "@type": "RelatedPlaceRefOrValue",
            "role": "Installation place",
            "place": {
              "@type": "PlaceRef",
              "@referredType": "GeographicSite",
              "id": "11",
              "href": "http://mycsp.com:8080/tmf-api/geographicSiteManagement/v5/geographicSite/11",
              "name": "Jean Pontus home"
            }
          }
        ],
        "@type": "Service"
      },
      "@type": "ServiceQualificationItem"
    },
    "serviceQualificationItem": [
      {
        "id": "1",
        "expirationDate": "2023-04-24T19:20:39.239Z",
        "service": {
          "serviceSpecification": {
            "id": "78",
            "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v5/serviceSpecification/78",
            "name": "X-Fiber Service",
            "@type": "ServiceSpecificationRef",
            "@referredType": "ServiceSpecification"
          },
          "@type": "Service"
        },
        "@type": "ServiceQualificationItem"
      }
    ],
    {

```

```

    "id": "2",
    "expirationDate": "2023-04-24T19:20:39.239Z",
    "service": {
      "serviceSpecification": {
        "id": "87",
        "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v5/serviceSpecification/87",
        "name": "Standard Fiber Service",
        "@type": "ServiceSpecificationRef",
        "referredType": "ServiceSpecification"
      },
      "@type": "Service"
    },
    "@type": "ServiceQualificationItem"
  }
],
"state": "acknowledged",
"@type": "QueryServiceQualification"
}

```

Updates partially a QueryServiceQualification

PATCH /queryServiceQualification/{id}?fields=...

Description

This operation allows partial updates of a QueryServiceQualification entity. Support of json/merge (<https://tools.ietf.org/html/rfc7396>) is mandatory, support of json/patch (<http://tools.ietf.org/html/rfc5789>) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
description	
effectiveQualificationDate	
errorMessage	
estimatedResponseDate	
expectedQualificationDate	
expirationDate	
externalIdentifier	
href	
id	
instantSyncQualification	
queryServiceQualificationDate	
relatedParty	

Non Patchable Attributes	Rule
searchCriteria	
serviceQualificationItem	
state	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
----------------------	------

Usage samples

Here's an example of a request for partial update of QueryServiceQualification using merge-patch+json with change on following attributes - externalIdentifier and expectedQualificationDate value.

Request

```
PATCH /queryServiceQualification/54
Content-Type: application/merge-patch+json
{
  "expectedQualificationDate": "2023-06-13T13:51:41.218Z",
  "externalIdentifier": [
    {
      "id": "CSQ1-001",
      "@type": "ExternalIdentifier"
    }
  ],
  "@type": "QueryServiceQualification"
}
```

Response

```
200
{
  "id": "54",
  "href": "http://mycsp.com:8080/tmf-api/serviceQualificationManagement/v5/queryServiceQualification/54",
  "description": "Query Service Qualification Illustration",
  "estimatedResponseDate": "2023-04-13T14:00:00.218Z",
  "expectedQualificationDate": "2023-06-13T13:51:41.218Z",
  "externalIdentifier": [
    {
      "id": "CSQ1-001",
      "@type": "ExternalIdentifier"
    }
  ],
  "instantSyncQualification": false,
  "queryServiceQualificationDate": "2023-04-13T13:51:41.218Z",
  "relatedParty": [
    {

```

```

    "role": "Buyer contact",
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "partyOrPartyRole": {
      "@type": "PartyRef",
      "id": "456",
      "href": "http://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
      "name": "Jean Pontus",
      "@referredType": "Individual"
    }
  },
  "searchCriteria": {
    "id": "1",
    "category": {
      "id": "1",
      "href": "http://mycsp.com:8080/tmf-api/serviceCatalogManagement/v5/category/1",
      "name": "Fiber Access",
      "@type": "ServiceCategoryRef",
      "@referredType": "ServiceCategory"
    },
    "service": {
      "place": [
        {
          "@type": "RelatedPlaceRefOrValue",
          "role": "Installation place",
          "place": {
            "@type": "PlaceRef",
            "@referredType": "GeographicSite",
            "id": "11",
            "href": "http://mycsp.com:8080/tmf-api/geographicSiteManagement/v5/geographicSite/11",
            "name": "Jean Pontus home"
          }
        }
      ],
      "@type": "Service"
    },
    "@type": "ServiceQualificationItem"
  },
  "state": "acknowledged",
  "@type": "QueryServiceQualification"
}

```

Here's an example of a request for partial update of QueryServiceQualification using json-patch+json with change on expectedQualificationDate value .

Request

```

PATCH /queryServiceQualification/54
Content-Type: application/json-patch+json
[
  {
    "op": "replace",
    "path": "expectedQualificationDate",
    "value": "2023-04-11T15:20:39.239Z"
  }
]

```

Response

```

200
{
  "id": "54",
  "href": "http://mycsp.com:8080/tmf-
api/serviceQualificationManagement/v5/queryServiceQualification/54",
  "description": "Query Service Qualification Illustration",
  "estimatedResponseDate": "2023-04-13T14:00:00.218Z",
  "expectedQualificationDate": "2023-06-13T13:51:41.218Z",
  "externalIdentifier": [
    {
      "id": "CSQ1-001",
      "@type": "ExternalIdentifier"
    }
  ],
  "instantSyncQualification": false,
  "queryServiceQualificationDate": "2023-04-13T13:51:41.218Z",
  "relatedParty": [
    {
      "role": "Buyer contact",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "id": "456",
        "href": "http://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
        "name": "Jean Pontus",
        "@referredType": "Individual"
      }
    }
  ],
  "searchCriteria": {
    "id": "1",
    "category": {
      "id": "1",
      "href": "http://mycsp.com:8080/tmf-
api/serviceCatalogManagement/v5/category/1",
      "name": "Fiber Access",
      "@type": "ServiceCategoryRef",
      "@referredType": "ServiceCategory"
    },
    "service": {
      "place": [
        {
          "@type": "RelatedPlaceRefOrValue",
          "role": "Installation place",
          "place": {
            "@type": "PlaceRef",
            "@referredType": "GeographicSite",
            "id": "11",
            "href": "http://mycsp.com:8080/tmf-
api/geographicSiteManagement/v5/geographicSite/11",
            "name": "Jean Pontus home"
          }
        }
      ],
      "@type": "Service"
    },
    "@type": "ServiceQualificationItem"
  },
  "state": "acknowledged",

```

```
"@type": "QueryServiceQualification"
}
```

Here's an example of a request for partial update of QueryServiceQualification using json-patch-query with change on relatedParty value.

Request

```
PATCH /queryServiceQualification/54
Content-Type: application/json-patch-query+json
[
  {
    "op": "replace",
    "path":
"/relatedParty/partyOrPartyRole/name?relatedParty.partyOrPartyRole.id=456",
    "value": "Ross"
  }
]
```

Response

```
200
{
  "id": "54",
  "href": "http://mycsp.com:8080/tmf-
api/serviceQualificationManagement/v5/queryServiceQualification/54",
  "description": "Query Service Qualification Illustration",
  "estimatedResponseDate": "2023-04-13T14:00:00.218Z",
  "expectedQualificationDate": "2023-06-13T13:51:41.218Z",
  "externalIdentifier": [
    {
      "id": "CSQ1-001",
      "@type": "ExternalIdentifier"
    }
  ],
  "instantSyncQualification": false,
  "queryServiceQualificationDate": "2023-04-13T13:51:41.218Z",
  "relatedParty": [
    {
      "role": "Buyer contact",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "id": "456",
        "href": "http://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
        "name": "Ross",
        "@referredType": "Individual"
      }
    }
  ],
  "searchCriteria": {
    "id": "1",
    "category": {
      "id": "1",
      "href": "http://mycsp.com:8080/tmf-
api/serviceCatalogManagement/v5/category/1",
      "name": "Fiber Access",
      "@type": "ServiceCategoryRef",

```

```

    "@referredType": "ServiceCategory"
  },
  "service": {
    "place": [
      {
        "@type": "RelatedPlaceRefOrValue",
        "role": "Installation place",
        "place": {
          "@type": "PlaceRef",
          "@referredType": "GeographicSite",
          "id": "11",
          "href": "http://mycsp.com:8080/tmf-
api/geographicSiteManagement/v5/geographicSite/11",
          "name": "Jean Pontus home"
        }
      }
    ],
    "@type": "Service"
  },
  "@type": "ServiceQualificationItem"
},
"state": "acknowledged",
"@type": "QueryServiceQualification"
}

```

Here's an example of a request for partial update of QueryServiceQualification with change on expectedQualificationDate, externalIdentifier, relatedParty value.

Request

```

PATCH /queryServiceQualification/54
Content-Type: application/json
{
  "expectedQualificationDate": "2023-06-13T13:51:41.218Z",
  "externalIdentifier": [
    {
      "id": "CSQ1-001",
      "@type": "ExternalIdentifier"
    }
  ],
  "@type": "QueryServiceQualification"
}

```

Response

```

200
{
  "id": "54",
  "href": "http://mycsp.com:8080/tmf-
api/serviceQualificationManagement/v5/queryServiceQualification/54",
  "description": "Query Service Qualification Illustration",
  "estimatedResponseDate": "2023-04-13T14:00:00.218Z",
  "expectedQualificationDate": "2023-06-13T13:51:41.218Z",
  "externalIdentifier": [
    {
      "id": "CSQ1-001",
      "@type": "ExternalIdentifier"
    }
  ]
}

```

```

    }
  ],
  "instantSyncQualification": false,
  "queryServiceQualificationDate": "2023-04-13T13:51:41.218Z",
  "relatedParty": [
    {
      "role": "Buyer contact",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "id": "490",
        "href": "http://mycsp.com:8080/tmf-api/partyManagement/v5/individual/490",
        "name": "Ross",
        "@referredType": "Individual"
      }
    }
  ],
  "searchCriteria": {
    "id": "1",
    "category": {
      "id": "1",
      "href": "http://mycsp.com:8080/tmf-api/serviceCatalogManagement/v5/category/1",
      "name": "Fiber Access",
      "@type": "ServiceCategoryRef",
      "@referredType": "ServiceCategory"
    },
    "service": {
      "place": [
        {
          "@type": "RelatedPlaceRefOrValue",
          "role": "Installation place",
          "place": {
            "@type": "PlaceRef",
            "@referredType": "GeographicSite",
            "id": "11",
            "href": "http://mycsp.com:8080/tmf-api/geographicSiteManagement/v5/geographicSite/11",
            "name": "Jean Pontus home"
          }
        }
      ],
      "@type": "Service"
    },
    "@type": "ServiceQualificationItem"
  },
  "state": "acknowledged",
  "@type": "QueryServiceQualification"
}

```

Deletes a QueryServiceQualification

DELETE /queryServiceQualification/{id}

Description

This operation deletes a QueryServiceQualification entity.

Usage samples

Here's an example of a request for deleting a specific QueryServiceQualification instance.

Request

```
DELETE /queryServiceQualification/54
Content-Type: application/json
```

Response

```
204
```

Operations on CheckServiceQualification

Retrieves a CheckServiceQualification by ID

GET /checkServiceQualification/{id}?fields=...

Description

This operation retrieves a CheckServiceQualification entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's a sample of a request for retrieving a CheckServiceQualification resource based on its id

Request

```
GET /checkServiceQualification/44
Content-Type: application/json
```

Response

```
200
{
  "id": "44",
  "href": "https://mycsp.com:8080/tmf-
api/serviceQualificationManagement/v5/checkServiceQualification/44",
  "checkServiceQualificationDate": "2023-04-10T19:20:39.239Z",
  "description": "check service qualification illustration",
  "effectiveQualificationDate": "2023-04-11T15:20:39.239Z",
  "estimatedResponseDate": "2023-04-11T15:20:39.239Z",
  "expectedQualificationDate": "2023-04-11T15:20:39.239Z",
  "expirationDate": "2023-04-24T19:20:39.239Z",
  "externalIdentifier": [
    {
      "id": "CSQ1",
      "@type": "ExternalIdentifier"
    }
  ]
}
```

```

    }
  ],
  "provideAlternative": true,
  "provideResultReason": true,
  "instantSyncQualification": false,
  "qualificationResult": "alternate",
  "relatedParty": [
    {
      "role": "Buyer contact",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "456",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
        "name": "Jean Pontus",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "serviceQualificationItem": [
    {
      "id": "1",
      "expectedActivationDate": "2023-04-16T19:20:39.239Z",
      "expectedServiceAvailabilityDate": "2023-04-11T19:20:39.239Z",
      "expirationDate": "2023-04-24T19:20:39.239Z",
      "qualificationResult": "alternate",
      "state": "done",
      "alternateServiceProposal": [
        {
          "id": "1",
          "alternateServiceAvailabilityDate": "2020-04-15T19:20:39.239Z",
          "alternateService": {
            "serviceCharacteristic": [
              {
                "id": "1",
                "name": "BroadcastQuality",
                "valueType": "string",
                "value": "HD",
                "@type": "StringCharacteristic"
              }
            ],
            "serviceSpecification": {
              "id": "23",
              "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v5/serviceSpecification/23",
              "name": "TVAccess",
              "@type": "ServiceSpecificationRef",
              "@referredType": "ServiceSpecification"
            },
            "@type": "Service"
          },
          "@type": "AlternateServiceProposal"
        }
      ],
      "eligibilityResultReason": [
        {
          "code": "26-001",
          "label": "not enough bandwidth at this place",
          "@type": "EligibilityResultReason"
        }
      ],
      "service": {
        "place": [

```

```

    {
      "role": "Installation place",
      "@type": "RelatedPlaceRefOrValue",
      "place": {
        "@type": "PlaceRef",
        "@referredType": "GeographicSite",
        "id": "11",
        "href": "https://mycsp.com:8080/tmf-
api/geographicSiteManagement/v5/geographicSite/11",
        "name": "Jean Pontus home"
      }
    }
  ],
  "serviceCharacteristic": [
    {
      "id": "1",
      "name": "BroadcastQuality",
      "valueType": "string",
      "value": "4K",
      "@type": "StringCharacteristic"
    }
  ],
  "serviceSpecification": {
    "id": "23",
    "href": "https://mycsp.com:8080/tmf-
api/serviceCatalogManagement/v5/serviceSpecification/23",
    "name": "TVAccess",
    "@type": "ServiceSpecificationRef",
    "@referredType": "ServiceSpecification"
  },
  "@type": "Service"
},
"@type": "CheckServiceQualificationItem"
}
],
"state": "done",
"@type": "CheckServiceQualification"
}

```

List or find CheckServiceQualification objects

GET /checkServiceQualification?fields=...

Description

This operation list CheckServiceQualification entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving all the CheckServiceQualification that were completed.

Request

```
GET /checkServiceQualification?filter=state=done&fields=id,name,state
```

Content-Type: application/json

Response

```

200
[
  {
    "id": "44",
    "href": "https://mycsp.com:8080/tmf-
api/serviceQualificationManagement/v5/checkServiceQualification/44",
    "checkServiceQualificationDate": "2023-04-10T19:20:39.239Z",
    "description": "check service qualification illustration",
    "effectiveQualificationDate": "2023-04-11T15:20:39.239Z",
    "estimatedResponseDate": "2023-04-11T15:20:39.239Z",
    "expectedQualificationDate": "2023-04-11T15:20:39.239Z",
    "expirationDate": "2023-04-24T19:20:39.239Z",
    "externalIdentifier": [
      {
        "id": "CSQ1",
        "@type": "ExternalIdentifier"
      }
    ],
    "provideAlternative": true,
    "provideResultReason": true,
    "instantSyncQualification": false,
    "qualificationResult": "alternate",
    "relatedParty": [
      {
        "role": "Buyer contact",
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "partyOrPartyRole": {
          "id": "456",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/456",
          "name": "Jean Pontus",
          "@type": "PartyRef",
          "@referredType": "Individual"
        }
      }
    ],
    "serviceQualificationItem": [
      {
        "id": "1",
        "expectedActivationDate": "2023-04-16T19:20:39.239Z",
        "expectedServiceAvailabilityDate": "2023-04-11T19:20:39.239Z",
        "expirationDate": "2023-04-24T19:20:39.239Z",
        "qualificationResult": "alternate",
        "state": "done",
        "alternateServiceProposal": [
          {
            "id": "1",
            "alternateServiceAvailabilityDate": "2020-04-15T19:20:39.239Z",
            "alternateService": {
              "serviceCharacteristic": [
                {
                  "id": "1",
                  "name": "BroadcastQuality",
                  "valueType": "string",
                  "value": "HD",
                  "@type": "StringCharacteristic"
                }
              ]
            }
          }
        ]
      }
    ]
  }
]

```

```

    }
  ],
  "serviceSpecification": {
    "id": "23",
    "href": "https://mycsp.com:8080/tmf-
api/serviceCatalogManagement/v5/serviceSpecification/23",
    "name": "TVAccess",
    "@type": "ServiceSpecificationRef",
    "@referredType": "ServiceSpecification"
  },
  "@type": "Service"
},
"@type": "AlternateServiceProposal"
}
],
"eligibilityResultReason": [
{
  "code": "26-001",
  "label": "not enough bandwidth at this place",
  "@type": "EligibilityResultReason"
}
],
"service": {
  "place": [
    {
      "role": "Installation place",
      "@type": "RelatedPlaceRefOrValue",
      "place": {
        "@type": "PlaceRef",
        "@referredType": "GeographicSite",
        "id": "11",
        "href": "https://mycsp.com:8080/tmf-
api/geographicSiteManagement/v5/geographicSite/11",
        "name": "Jean Pontus home"
      }
    }
  ],
  "serviceCharacteristic": [
    {
      "id": "1",
      "name": "BroadcastQuality",
      "valueType": "string",
      "value": "4K",
      "@type": "StringCharacteristic"
    }
  ],
  "serviceSpecification": {
    "id": "23",
    "href": "https://mycsp.com:8080/tmf-
api/serviceCatalogManagement/v5/serviceSpecification/23",
    "name": "TVAccess",
    "@type": "ServiceSpecificationRef",
    "@referredType": "ServiceSpecification"
  },
  "@type": "Service"
},
"@type": "CheckServiceQualificationItem"
}
],
"state": "done",
"@type": "CheckServiceQualification"
}

```

]

Here's an example of a request for retrieving all the CheckServiceQualification with field selection.

Request

```
GET
/checkServiceQualification?fields=id,description,effectiveQualificationDate,expirationDate,qualificationResult&filter=state=done
Content-Type: application/json
```

Response

```
200
[
  {
    "id": "48",
    "href": "https://mycsp.com:8080/tmf-api/serviceQualificationManagement/v5/checkServiceQualification/48",
    "description": "check service qualification for IP",
    "effectiveQualificationDate": "2023-04-11T14:23:33.239Z",
    "expirationDate": "2023-04-11T14:23:33.239Z",
    "qualificationResult": "qualified",
    "@type": "CheckServiceQualification"
  },
  {
    "id": "44",
    "href": "https://mycsp.com:8080/tmf-api/serviceQualificationManagement/v5/checkServiceQualification/44",
    "description": "check service qualification illustration",
    "effectiveQualificationDate": "2023-04-11T15:20:39.239Z",
    "expirationDate": "2023-04-11T14:23:33.239Z",
    "qualificationResult": "alternate",
    "@type": "CheckServiceQualification"
  },
  {
    "id": "41",
    "href": "https://mycsp.com:8080/tmf-api/serviceQualificationManagement/v5/checkServiceQualification/41",
    "description": "check service qualification for an UNI access",
    "effectiveQualificationDate": "2023-04-11T17:10:39.239Z",
    "expirationDate": "2023-04-11T14:23:33.239Z",
    "qualificationResult": "qualified",
    "@type": "CheckServiceQualification"
  }
]
```

Initiate a task for CheckServiceQualification

POST /checkServiceQualification?fields=...

Description

This operation creates a CheckServiceQualification entity.

Mandatory Attributes

Mandatory Attributes	Rule
serviceQualificationItem	
@type	M

Usage samples

Here's an example of a request for creating CheckServiceQualification.

Request

```
POST /checkServiceQualification
Content-Type: application/json
{
  "description": "check service qualification illustration",
  "expectedQualificationDate": "2023-04-11T15:20:39.239Z",
  "externalIdentifier": [
    {
      "id": "CSQ1",
      "@type": "ExternalIdentifier"
    }
  ],
  "provideAlternative": true,
  "instantSyncQualification": false,
  "provideResultReason": true,
  "relatedParty": [
    {
      "role": "Buyer contact",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "456",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
        "name": "Jean Pontus",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "serviceQualificationItem": [
    {
      "id": "1",
      "expectedServiceAvailabilityDate": "2023-04-11T19:20:39.239Z",
      "service": {
        "place": [
          {
            "role": "Installation place",
            "@type": "RelatedPlaceRefOrValue",
            "place": {
              "@type": "PlaceRef",
              "@referredType": "GeographicSite",
              "id": "11",
              "href": "https://mycsp.com:8080/tmf-api/geographicSiteManagement/v5/geographicSite/11",
              "name": "Jean Pontus home"
            }
          }
        ]
      }
    }
  ],
}
```

```

    "serviceCharacteristic": [
      {
        "id": "1",
        "name": "BroadcastQuality",
        "valueType": "string",
        "value": "4K",
        "@type": "StringCharacteristic"
      }
    ],
    "@type": "Service"
  },
  "@type": "CheckServiceQualificationItem"
}
],
"@type": "CheckServiceQualification"
}

```

Response

```

201
{
  "id": "44",
  "href": "https://mycsp.com:8080/tmf-api/serviceQualificationManagement/v5/checkServiceQualification/44",
  "checkServiceQualificationDate": "2023-04-10T19:20:39.239Z",
  "description": "check service qualification illustration",
  "effectiveQualificationDate": "2023-04-11T15:20:39.239Z",
  "estimatedResponseDate": "2023-04-11T15:20:39.239Z",
  "expectedQualificationDate": "2023-04-11T15:20:39.239Z",
  "expirationDate": "2023-04-24T19:20:39.239Z",
  "externalIdentifier": [
    {
      "id": "CSQ1",
      "@type": "ExternalIdentifier"
    }
  ],
  "provideAlternative": true,
  "provideResultReason": true,
  "instantSyncQualification": false,
  "qualificationResult": "alternate",
  "relatedParty": [
    {
      "role": "Buyer contact",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "456",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
        "name": "Jean Pontus",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "serviceQualificationItem": [
    {
      "id": "1",
      "expectedActivationDate": "2023-04-16T19:20:39.239Z",
      "expectedServiceAvailabilityDate": "2023-04-11T19:20:39.239Z",
      "expirationDate": "2023-04-24T19:20:39.239Z",
      "qualificationResult": "alternate",
    }
  ]
}

```

```

"state": "done",
"alternateServiceProposal": [
  {
    "id": "1",
    "alternateServiceAvailabilityDate": "2023-04-15T19:20:39.239Z",
    "alternateService": {
      "serviceCharacteristic": [
        {
          "id": "1",
          "name": "BroadcastQuality",
          "valueType": "string",
          "value": "HD",
          "@type": "StringCharacteristic"
        }
      ],
      "serviceSpecification": {
        "id": "23",
        "href": "https://mycsp.com:8080/tmf-
api/serviceCatalogManagement/v5/serviceSpecification/23",
        "name": "TVAccess",
        "@type": "ServiceSpecificationRef",
        "@referredType": "ServiceSpecification"
      },
      "@type": "Service"
    },
    "@type": "AlternateServiceProposal"
  }
],
"eligibilityResultReason": [
  {
    "code": "26-001",
    "label": "not enough bandwidth at this place",
    "@type": "EligibilityResultReason"
  }
],
"service": {
  "place": [
    {
      "role": "Installation place",
      "@type": "RelatedPlaceRefOrValue",
      "place": {
        "@type": "PlaceRef",
        "@referredType": "GeographicSite",
        "id": "11",
        "href": "https://mycsp.com:8080/tmf-
api/geographicSiteManagement/v5/geographicSite/11",
        "name": "Jean Pontus home"
      }
    }
  ],
  "serviceCharacteristic": [
    {
      "id": "1",
      "name": "BroadcastQuality",
      "valueType": "string",
      "value": "4K",
      "@type": "StringCharacteristic"
    }
  ],
  "@type": "Service"
},
"@type": "CheckServiceQualificationItem"
}

```

```

    ],
    "state": "done",
    "@type": "CheckServiceQualification"
  }

```

Updates partially a CheckServiceQualification

PATCH /checkServiceQualification/{id}?fields=...

Description

This operation allows partial updates of a CheckServiceQualification entity. Support of json/merge (<https://tools.ietf.org/html/rfc7396>) is mandatory, support of json/patch (<http://tools.ietf.org/html/rfc5789>) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
checkServiceQualificationDate	
description	
effectiveQualificationDate	
errorMessage	
estimatedResponseDate	
expectedQualificationDate	
expirationDate	
externalIdentifier	
href	
id	
instantSyncQualification	
provideAlternative	
provideResultReason	
qualificationResult	
relatedParty	
serviceQualificationItem	
state	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
----------------------	------

Usage samples

Here's an example of a request for patching a CheckServiceQualification resource (created via previous POST) with changes on attributes- externalIdentifier and expectedQualificationDate using merge-patch+json.

Request

```
PATCH /checkServiceQualification/44
Content-Type: application/merge-patch+json
{
  "expectedQualificationDate": "2023-06-15T15:20:39.239Z",
  "externalIdentifier": [
    {
      "id": "CSQ1-001",
      "@type": "ExternalIdentifier"
    }
  ],
  "@type": "CheckServiceQualification"
}
```

Response

```
200
{
  "id": "44",
  "href": "https://mycsp.com:8080/tmf-api/serviceQualificationManagement/v5/checkServiceQualification/44",
  "checkServiceQualificationDate": "2023-04-10T19:20:39.239Z",
  "description": "check service qualification illustration",
  "expectedQualificationDate": "2023-06-15T15:20:39.239Z",
  "externalIdentifier": [
    {
      "id": "CSQ1-001",
      "@type": "ExternalIdentifier"
    }
  ],
  "provideAlternative": true,
  "provideResultReason": true,
  "instantSyncQualification": false,
  "qualificationResult": "unqualified",
  "relatedParty": [
    {
      "role": "Buyer contact",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "id": "456",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
        "name": "Jean Pontus",
        "@referredType": "Individual"
      }
    }
  ],
  "serviceQualificationItem": [
```

```

{
  "id": "1",
  "expectedServiceAvailabilityDate": "2023-04-11T19:20:39.239Z",
  "qualificationResult": "unqualified",
  "state": "acknowledged",
  "service": {
    "place": [
      {
        "role": "Installation place",
        "@type": "RelatedPlaceRefOrValue",
        "place": {
          "@type": "PlaceRef",
          "@referredType": "GeographicSite",
          "id": "11",
          "href": "https://mycsp.com:8080/tmf-api/geographicSiteManagement/v5/geographicSite/11",
          "name": "Jean Pontus home"
        }
      }
    ],
    "serviceCharacteristic": [
      {
        "id": "1",
        "name": "BroadcastQuality",
        "valueType": "string",
        "value": "4K",
        "@type": "StringCharacteristic"
      }
    ],
    "serviceSpecification": {
      "id": "23",
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v5/serviceSpecification/23",
      "name": "TVAccess",
      "@type": "ServiceSpecificationRef",
      "@referredType": "ServiceSpecification"
    },
    "@type": "Service"
  },
  "@type": "CheckServiceQualificationItem"
},
"state": "acknowledged",
"@type": "CheckServiceQualification"
}

```

Here's an example of a request for patching a CheckServiceQualification resource (created via previous POST) with updated expectedQualificationDate using json-patch+json.

Request

```

PATCH /checkServiceQualification/44
Content-Type: application/json-patch+json
[
  {
    "op": "replace",
    "path": "expectedQualificationDate",
    "value": "2023-04-11T15:20:39.239Z"
  }
]

```

Response

```

200
{
  "id": "44",
  "href": "https://mycsp.com:8080/tmf-
api/serviceQualificationManagement/v5/checkServiceQualification/44",
  "checkServiceQualificationDate": "2023-04-10T19:20:39.239Z",
  "description": "check service qualification illustration",
  "expectedQualificationDate": "2023-04-11T15:20:39.239Z",
  "externalIdentifier": [
    {
      "id": "CSQ1-001",
      "@type": "ExternalIdentifier"
    }
  ],
  "provideAlternative": true,
  "provideResultReason": true,
  "instantSyncQualification": false,
  "qualificationResult": "unqualified",
  "relatedParty": [
    {
      "role": "Buyer contact",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "id": "456",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
        "name": "Jean Pontus",
        "@referredType": "Individual"
      }
    }
  ],
  "serviceQualificationItem": [
    {
      "id": "1",
      "expectedServiceAvailabilityDate": "2023-04-11T19:20:39.239Z",
      "qualificationResult": "unqualified",
      "state": "acknowledged",
      "service": {
        "place": [
          {
            "role": "Installation place",
            "@type": "RelatedPlaceRefOrValue",
            "place": {
              "@type": "PlaceRef",
              "@referredType": "GeographicSite",
              "id": "11",
              "href": "https://mycsp.com:8080/tmf-
api/geographicSiteManagement/v5/geographicSite/11",
              "name": "Jean Pontus home"
            }
          }
        ]
      }
    }
  ],
  "serviceCharacteristic": [
    {
      "id": "1",
      "name": "BroadcastQuality",
      "valueType": "string",
      "value": "4K",
      "@type": "StringCharacteristic"
    }
  ]
}

```

```

    }
  ],
  "serviceSpecification": {
    "id": "23",
    "href": "https://mycsp.com:8080/tmf-
api/serviceCatalogManagement/v5/serviceSpecification/23",
    "name": "TVAccess",
    "@type": "ServiceSpecificationRef",
    "@referredType": "ServiceSpecification"
  },
  "@type": "Service"
},
"@type": "CheckServiceQualificationItem"
}
],
"state": "acknowledged",
"@type": "CheckServiceQualification"
}

```

Here's an example of a request for patching a CheckServiceQualification resource (created via previous POST) with change on expectedActivationDate using json-patch-query.

Request

```

PATCH /checkServiceQualification/44
Content-Type: application/json-patch-query+json
[
  {
    "op": "replace",
    "path":
"/serviceQualificationItem/expectedActivationDate?serviceQualificationItem.id=1",
    "value": "2023-04-16T19:20:39.239Z"
  }
]

```

Response

```

200
{
  "id": "44",
  "href": "https://mycsp.com:8080/tmf-
api/serviceQualificationManagement/v5/checkServiceQualification/44",
  "checkServiceQualificationDate": "2023-04-10T19:20:39.239Z",
  "description": "check service qualification illustration",
  "expectedQualificationDate": "2023-06-15T15:20:39.239Z",
  "externalIdentifier": [
    {
      "id": "CSQ1-001",
      "@type": "ExternalIdentifier"
    }
  ],
  "provideAlternative": true,
  "provideResultReason": true,
  "instantSyncQualification": false,
  "qualificationResult": "unqualified",
  "relatedParty": [
    {
      "role": "Buyer contact",

```

```

    "@type": "RelatedPartyRefOrPartyRoleRef",
    "partyOrPartyRole": {
      "@type": "PartyRef",
      "id": "456",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
      "name": "Jean Pontus",
      "@referredType": "Individual"
    }
  },
  "serviceQualificationItem": [
    {
      "id": "1",
      "expectedActivationDate": "2023-04-16T19:20:39.239Z",
      "expectedServiceAvailabilityDate": "2023-04-11T19:20:39.239Z",
      "qualificationResult": "unqualified",
      "state": "acknowledged",
      "service": {
        "place": [
          {
            "role": "Installation place",
            "@type": "RelatedPlaceRefOrValue",
            "place": {
              "@type": "PlaceRef",
              "@referredType": "GeographicSite",
              "id": "11",
              "href": "https://mycsp.com:8080/tmf-api/geographicSiteManagement/v5/geographicSite/11",
              "name": "Jean Pontus home"
            }
          }
        ]
      }
    },
    {
      "id": "1",
      "name": "BroadcastQuality",
      "valueType": "string",
      "value": "4K",
      "@type": "StringCharacteristic"
    }
  ],
  "serviceSpecification": {
    "id": "23",
    "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v5/serviceSpecification/23",
    "name": "TVAccess",
    "@type": "ServiceSpecificationRef",
    "@referredType": "ServiceSpecification"
  },
  "@type": "Service"
},
"@type": "CheckServiceQualificationItem"
},
"state": "acknowledged",
"@type": "CheckServiceQualification"
}

```

Here's an example of a request for patching a CheckServiceQualification resource (created via previous POST) with change on attributes- externalIdentifier and expectedQualificationDate value

Request

```
PATCH /checkServiceQualification/44
Content-Type: application/json
{
  "expectedQualificationDate": "2023-06-15T15:20:39.239Z",
  "externalIdentifier": [
    {
      "id": "CSQ1-001",
      "@type": "ExternalIdentifier"
    }
  ],
  "@type": "CheckServiceQualification"
}
```

Response

```
200
{
  "id": "44",
  "href": "https://mycsp.com:8080/tmf-api/serviceQualificationManagement/v5/checkServiceQualification/44",
  "checkServiceQualificationDate": "2023-04-10T19:20:39.239Z",
  "description": "check service qualification illustration",
  "expectedQualificationDate": "2023-06-15T15:20:39.239Z",
  "externalIdentifier": [
    {
      "id": "CSQ1-001",
      "@type": "ExternalIdentifier"
    }
  ],
  "provideAlternative": true,
  "provideResultReason": true,
  "instantSyncQualification": false,
  "qualificationResult": "unqualified",
  "relatedParty": [
    {
      "role": "Buyer contact",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "id": "456",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/456",
        "name": "Jean Pontus",
        "@referredType": "Individual"
      }
    }
  ],
  "serviceQualificationItem": [
    {
      "id": "1",
      "expectedServiceAvailabilityDate": "2023-04-11T19:20:39.239Z",
      "qualificationResult": "unqualified",
      "state": "acknowledged",
      "service": {
        "place": [
          {
            "role": "Installation place",
            "@type": "RelatedPlaceRefOrValue",

```

```

        "place": {
            "@type": "PlaceRef",
            "@referredType": "GeographicSite",
            "id": "11",
            "href": "https://mycsp.com:8080/tmf-
api/geographicSiteManagement/v5/geographicSite/11",
            "name": "Jean Pontus home"
        }
    },
    "serviceCharacteristic": [
        {
            "id": "1",
            "name": "BroadcastQuality",
            "valueType": "string",
            "value": "4K",
            "@type": "StringCharacteristic"
        }
    ],
    "serviceSpecification": {
        "id": "23",
        "href": "https://mycsp.com:8080/tmf-
api/serviceCatalogManagement/v5/serviceSpecification/23",
        "name": "TVAccess",
        "@type": "ServiceSpecificationRef",
        "@referredType": "ServiceSpecification"
    },
    "@type": "Service"
},
"@type": "CheckServiceQualificationItem"
}
],
"state": "acknowledged",
"@type": "CheckServiceQualification"
}

```

Deletes a CheckServiceQualification

DELETE /checkServiceQualification/{id}

Description

This operation deletes a CheckServiceQualification entity.

Usage samples

Here's an example of a request for deleting a specific CheckServiceQualification instance.

Request

```

DELETE /checkServiceQualification/456
Content-Type: application/json

```

Response

```

204

```

```
{  
  "id": "456",  
  "href": "https://mycsp.com:8080/tmf-  
api/serviceQualificationManagement/v5/checkServiceQualification/456",  
  "@type": "CheckServiceQualification"  
}
```

API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request

```
POST /api/hub

Accept: application/json

{
  "callback": "http://in.listener.com"
}
```

Response

```
201

Content-Type: application/json

Location: /api/hub/42

{
  "id": "42",
  "callback": "http://in.listener.com",
  "query": ""
}
```

Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request

```
DELETE /api/hub/42
Accept: application/json
```

Response

```
204
```

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

```
POST /client/listener

Accept: application/json

{
  "event": {
    EVENT BODY
  },
  "eventType": "EVENT_TYPE"
}
```

Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

Acknowledgements

Release History

Release Number	Date	Release led by:	Description
Version 5.0.0	June 2025	Amita Giriya Telstra amita.giriya@team.telstra.com Johnson Drago Telstra johnson.drago@team.telstra.com	Upgrade to V5 design patterns and OAS3

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